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Fractured Adelic Geometry

A Foundational Framework for the Birch and Swinnerton-Dyer Conjecture

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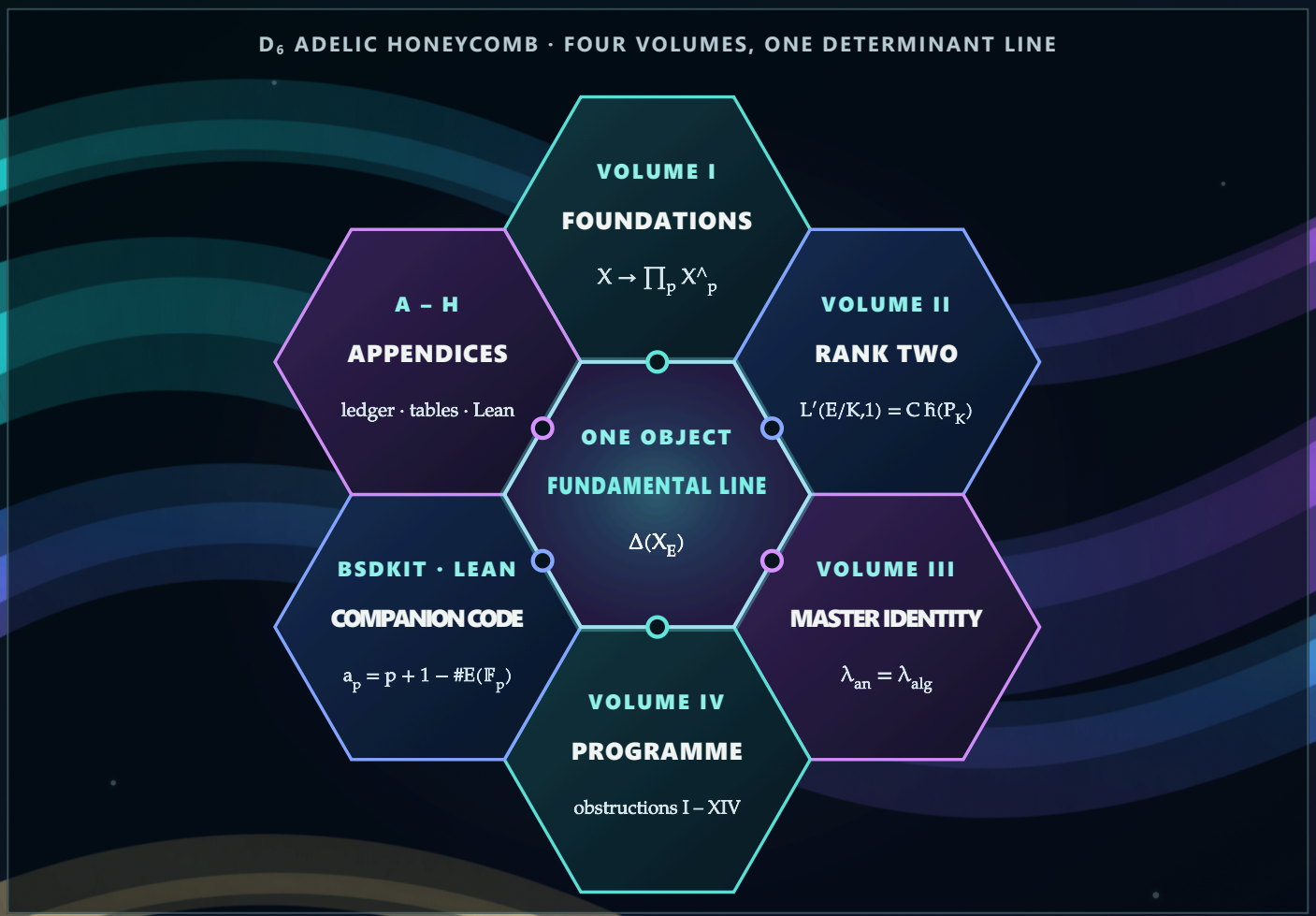

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English · 2026

Fractured Adelic Geometry

The Complete Monograph

Fractured Adelic Geometry

A Foundational Framework for the Birch and Swinnerton-Dyer Conjecture

Frobenius–Arakelov–Prismatic Geometry,
Higher Euler Systems, and the Master Identity

with a complete computational architecture
and a formal proof-obligation ledger

Vincent Liu

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The Birch and Swinnerton-Dyer conjecture remains open. *The Birch and Swinnerton-Dyer conjecture is, at the time of writing, an open problem. This book does not prove it, and no statement in it should be read as claiming otherwise.* What this book offers is a foundational reorganisation of the subject, a precisely axiomatised framework in which the conjecture becomes a single determinant identity rather than a comparison of two unrelated quantities, a complete inventory of exactly what would remain to be proved in that framework, and an implemented computational and formal-verification architecture for attacking, testing, and certifying the remaining statements. Every non-trivial assertion carries an explicit epistemic tag: [KNOWN], [CONDITIONAL], [NEW], or [OPEN]. The conventions governing these tags are set out in the section *Statement of Epistemic Status*, which the reader is asked to read before Chapter 1.

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To those who insisted that the right definition
is worth more than a thousand computations,

and to those who insisted on the computations.

Series Note

Fractured Adelic Geometry is the fourth work in the WiseMotif Monograph of Mathematical Intelligence and belongs to the Grothendieckian Programme. The first two works of the series are bound closely together. *The Grothendieckian Machine* set out how mathematical worlds are constructed, compared and carried from one setting to another, and how a machine may hold them as typed, versioned and verifiable objects. *GM-Omega-Infinity: Topos-Native Intelligence* stated what an intelligence must be, internally, if it is to reason inside such worlds. The third work, *Arithmetic Spectral Descent*, took the Riemann Hypothesis as a research subject of the Programme. The present work is a self-contained system in the same sense, and it is the Programme's second research subject.

The Programme has a single research aim: machine intelligence that discovers mathematical structure on its own account. That aim asks for reasoning architectures held to the standard of proof, and for a discipline of recursive mathematical discovery, in which a proposed structure is tested, what survives the test is abstracted into a new setting, and the question is posed again in that setting. Its model is Grothendieck's rising sea. A hard problem is not attacked where it stands; a theory is raised around it until the problem becomes natural and gives way. A research subject of the Programme is a problem on which this method is carried through in full, with every step recorded in a form that a mathematician can check and a machine can execute.

This work carries it through on the conjecture of Birch and Swinnerton-Dyer. It begins from a diagnosis: every proved case of the conjecture crosses from the analytic side to the arithmetic side by a bridge built for one point and one direction, and the bridges stop at analytic rank one. The recursion therefore changes the object rather than the bridge. It proposes a category of Frobenius–Arakelov–Prismatic spaces governed by twelve axioms, attaches to an elliptic curve a single arithmetic spectrum carrying one determinant line, and restates the conjecture as the agreement of the line's two trivialisations. Each step of that construction is then tested against the cases in which the answer is known — curves over finite fields, $\mathbb{G}_m$ over $\mathbb{Z}$, analytic rank at most one — and what fails is recorded as an obstruction. The recursion ends in a register of twelve conjectures with stated difficulty, an atlas of fourteen obstructions with attack plans, a computational and formal-verification architecture, and a staged construction programme with explicit criteria for abandonment.

The work keeps one editorial principle throughout. No result is presented at a strength greater than its proof. Every substantive assertion carries one of four labels — **[KNOWN]**, **[CONDITIONAL]**, **[NEW]**, **[OPEN]** — where it is made, and every conditional implication is printed with its hypotheses. Computation is reported as computation, at the level of certification it has, and is never promoted to a theorem. An obstruction the work fails to remove is stated as an obstruction, in the place where it stops the argument.

The Birch and Swinnerton-Dyer conjecture remains open. This work does not prove it, proves no case of it that was not already known, and proves the finiteness of Ш for no curve. What it offers is the state of the question after the method has been applied without concession: what the framework would deliver, what it would need, and where each route to it stops.

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Abstract

This monograph asks why sixty years of work on the conjecture of Birch and Swinnerton-Dyer have produced theorems only in analytic rank at most one, and what kind of object would make the conjecture a single statement instead of a comparison between two unrelated quantities.

[KNOWN] The diagnosis is structural. Every proved case crosses from the analytic side to the arithmetic side by a bridge — Heegner points, Kolyvagin systems, Kato's Euler system — that carries one class in one direction. Chapter 27 isolates the rank-two obstruction as a statement about the exterior algebra acting on a derived deformation ring, not as a technical accident.

[NEW] The proposal changes the object. A category of Frobenius–Arakelov–Prismatic (FAP) spaces carries condensed analytic structure, integral prismatic F-gauges at every prime, Fargues–Fontaine realisations, an Arakelov boundary, an absolute cyclotomic structure and an arithmetic flow as foundational data, governed by twelve axioms F1–F12. To an elliptic curve $E/\mathbb{Q}$ it attaches an arithmetic spectrum X_E, its cohomology, a categorified Lefschetz–Verdier trace formula and a fundamental line $\Delta(X_E)$. The conjecture becomes the Master Identity (Conjecture A): the analytic and algebraic trivialisations of $\Delta(X_E)$ agree integrally.

[CONDITIONAL] The descent is proved: from Conjecture A and the axioms, the rank formula and the full leading-coefficient formula follow (Chapters 38–39). In the degenerate cases the framework reproduces known theorems: over a finite field nine of the twelve axioms become theorems and the formalism specialises to the theorem of Tate and Milne; for $\mathbb{G}_m$ over $\mathbb{Z}$ it specialises to the explicit formula and the class number formula (Chapter 40).

[OPEN] What remains is stated as a register and an atlas. Twelve conjectures carry the programme; eleven are rated at the two highest levels of difficulty, and only one is tractable with present technique. Fourteen obstructions stand between the framework and the conjecture — eight to constructing the framework, six on the proposed routes — and five of them contain a known major open problem: Conjecture A is the equivariant Tamagawa number conjecture for $h^1(E)(1)$ in the notation of this book, and the Selmer fixed-point principle contains the finiteness of Ш.

[KNOWN] The computational and verification architecture is specified in full: a seven-layer system with certified arithmetic, descent, L-function and p-adic engines, a certificate hierarchy C1–C7 with a stated trust model, a proof-obligation ledger, and a Lean 4 development of the statements, published uncompiled. Every numerical value is reported with its level of certification.

The Birch and Swinnerton-Dyer conjecture remains open. This monograph does not prove it, proves no case of it that was not already known, and proves the finiteness of Ш for no curve.

Preface

The question and its boundary

The conjecture of Birch and Swinnerton-Dyer predicts that two invariants of an elliptic curve over $\mathbb{Q}$, computed by mechanisms that share nothing, are equal: the order of vanishing of an analytic function at a point outside its region of convergence, and the rank of a finitely generated abelian group; and, at the next order, the leading coefficient of that function and a product of arithmetic invariants of which one, the Tate–Shafarevich group, is not known to be finite. This book asks what kind of object would make these one statement rather than two, and it records, step by step, how far that question can be answered today.

The boundary of the work is stated before anything else. The Birch and Swinnerton-Dyer conjecture remains open. This book does not prove it, proves no case of it that was not already known, and proves the finiteness of Ш for no curve. Every assertion in it carries one of four labels — [KNOWN], [CONDITIONAL], [NEW], [OPEN] — at the place where it is made, and the conventions governing them are set out in the *Statement of Epistemic Status*, which the reader is asked to read before Chapter 1.

A research subject of the Grothendieckian Programme

This work belongs to the Grothendieckian Programme of the WiseMotif Monograph of Mathematical Intelligence, and it is one of the Programme's research subjects. The Programme has a single aim: machine intelligence that discovers mathematical structure on its own account, under reasoning architectures held to the standard of proof. A research subject is a problem on which that aim is tested in full. The test is not whether a machine can produce text about a famous problem; it is whether the method can raise a structure around the problem, check that structure against everything already known, and say precisely where it fails.

The method is recursive mathematical discovery. A structure is proposed; it is tested against the cases in which the answer is known; what survives is abstracted into a new setting, and the question is posed again there. What does not survive is not discarded. It becomes an obstruction, with a statement, a status and an attack plan, and it is carried forward as part of the result. The book is the record of one such recursion, carried out on the arithmetic of elliptic curves, and the obstructions it records are as much its content as the constructions.

The rising sea

Grothendieck described two ways of opening a hard nut: to strike it until it breaks, or to immerse it in a rising sea until it softens and opens of itself. The proved cases of the conjecture of Birch and Swinnerton-Dyer are, almost without exception, of the first kind. Gross and Zagier compute two numbers and find them equal; Kolyvagin builds a system of cohomology classes and bounds a Selmer group; Kato builds an Euler system; Skinner and Urban prove a main conjecture. Each is a precise blow at one point, and the point is always analytic rank at most one.

This book tries the second way. It does not look for a sharper blow at rank two. It asks what the space would have to be on which the two sides of the conjecture are two descriptions of one object, writes down the axioms such a space must satisfy, and follows their consequences. The sea has not yet risen far enough: the object is specified, not constructed, and the specification rests on conjectures of the first rank of difficulty. But the specification is exact, it reproduces every degenerate case in which the answer is known, and it locates every remaining difficulty at a named place. That is the state the book reports.

What this book is about

In 1965 Birch and Swinnerton-Dyer, having computed on the EDSAC II at Cambridge the number of points modulo p on a great many elliptic curves, observed that the quantity

$$\prod_{p \le X} \frac{N_p}{p}$$

appeared to grow like $(\log X)^r$, where r was the rank of the group of rational points. From that observation grew the conjecture which now bears their names, and which asserts, in its refined form, that for an elliptic curve $E/\mathbb{Q}$

$$\begin{aligned} \operatorname{ord}_{s=1} L(E,s) &= \operatorname{rank} E(\mathbb{Q}), \\ \lim_{s \to 1} \frac{L(E,s)}{(s-1)^r} &= \frac{\#Ш(E/\mathbb{Q}) \cdot \operatorname{Reg}(E/\mathbb{Q}) \cdot \Omega_E \cdot \prod_p c_p}{\left(\#E(\mathbb{Q})_{\mathrm{tors}}\right)^2}. \end{aligned}$$

Sixty years later the conjecture is open. It is open in a very particular way, and the shape of that openness is the subject of this book.

The two sides of the conjecture do not merely resist comparison; they do not live in the same mathematical universe. On the left stands an analytic function of a complex variable, defined by an Euler product whose local factors encode point counts over finite fields, continued meromorphically by an appeal to modularity, and evaluated at a point outside the region of convergence. On the right stands a finitely generated abelian group, together with a collection of arithmetic invariants—a regulator built from an archimedean height pairing, a mysterious finite group Ш whose finiteness is itself unproved, local Tamagawa factors, a period, a torsion subgroup—each computed by an entirely different mechanism. There is no map in either direction. Every known proof of a fragment of the conjecture proceeds by constructing, with enormous ingenuity, a *bridge*: Heegner points, Kolyvagin's derived classes, Kato's Euler system, Beilinson–Flach elements. Each bridge is a miracle, and each miracle is local: it carries information across the chasm at exactly one point, and it carries it in exactly one direction.

The thesis of this book is that the bridges are a symptom, not a solution. They are needed because the foundations of arithmetic geometry, as bequeathed to us by Grothendieck's *Éléments* and *Séminaires*, describe an arithmetic scheme as a locally ringed space over $\operatorname{Spec}\mathbb{Z}$ — an object that sees Zariski localisation and étale descent superbly, and sees Frobenius dynamics, archimedean completion, integral p-adic Hodge structures, and regularised determinants not at all. In that language the two sides of BSD are different kinds of object, and a comparison between them can only ever be a bridge.

We propose instead to change the object.

What is proposed

The central construction of this book is a category of spaces — we call them *Frobenius–Arakelov–Prismatic spaces*, or FAP spaces — in which an arithmetic scheme is not a space over $\mathrm{Spec}\mathbb{Z}$ but a *cohesive family of its realisations together with the transition laws between arithmetic time scales*. An FAP space carries, simultaneously and as foundational rather than auxiliary data:

(i) a spectral and condensed analytic structure sheaf, in the sense of Clausen–Scholze and Lurie;
(ii) integral prismatic F-gauge structures at every finite place, in the sense of Bhatt–Scholze and Bhatt–Lurie;
(iii) a v-stack generic fibre with Fargues–Fontaine realisations;
(iv) an Arakelov boundary at the real place;
(v) an absolute cyclotomic (Λ-) structure encoding all Frobenius lifts;
(vi) a continuous arithmetic flow whose closed orbits are the finite places;
(vii) a determinant theory for nuclear objects, compatible with an integral determinant lattice.

Twelve axioms, F1–F12, pin down what such an object must satisfy. From them we extract an object X_E attached to an elliptic curve $E/\mathbb{Q}$, its cohomology $\mathrm{R}\Gamma_{\mathrm{FAP}}(X_E, -)$, a Lefschetz–Verdier trace formula, a Grothendieck–Riemann–Roch theorem, and finally a single statement — the *Master Identity* — which asserts that two trivialisations of one determinant line agree. The order of vanishing of $L(E, s)$ at $s = 1$ becomes the dimension of a kernel; the leading coefficient becomes the determinant of the induced pairing; the finiteness of Ш becomes a constructibility statement. The comparison disappears, because there is nothing left to compare.

That is the mathematical proposal. It is stated precisely in Part VI, and its consequences for the classical conjecture are derived in Chapters 38–40.

What is *not* proposed

We do not prove the Birch and Swinnerton-Dyer conjecture. We wish to say this as plainly as it can be said, at the front of the book, and we repeat it in the section *Statement of Epistemic Status* which follows this preface.

The Master Identity is a conjecture. Its ingredients rest on a construction problem — the existence of the adelic structured ∞-topos of Chapter 7 — which we formulate precisely, reduce to eight named obstructions, and do not solve. Several of those obstructions are, on any realistic assessment, of a difficulty comparable to the conjecture itself. Obstruction IV, the construction of the arithmetic flow, contains within it the substance of Deninger's programme; Obstruction VII, the identification of the Selmer fixed locus, contains within it the finiteness of Ш.

What we claim is different, and we believe it is worth claiming. We claim that the framework is *correct in its degenerate cases*: over a function field it specialises to the Grothendieck–Lefschetz trace formula and reproduces the theorem of Tate and Milne; for $\mathbb{G}_m$ over $\mathbb{Z}$ it specialises to the Riemann–Weil explicit formula and the class number formula; at $s = 0$ and for rank 0 curves it specialises to statements proved by Kolyvagin, Kato, Skinner–Urban, and Wei Zhang. We claim that it *explains* why the existing proofs stop where they stop: Chapter 27 isolates the rank-two obstruction as a precise structural statement about the exterior algebra acting on a derived deformation ring, rather than as a technical accident. We claim that it *predicts*: Chapters 35 and 36 extract numerical predictions which are falsifiable with the engine described in Part VII, and Chapter 55 explains what would refute the framework. And we

claim that it is *buildable*: Parts VII and VIII describe, and the accompanying source distribution implements, a computational and formal-verification architecture in which every claim in this book is a row in a proof-obligation ledger with an explicit status.

On being more radical than EGA

It is not a small thing to propose a replacement for the foundations that Grothendieck gave us, and the reader is entitled to ask on what grounds we think it justified. The grounds are these.

EGA and SGA solved a specific foundational problem: how to do algebraic geometry over an arbitrary base, with nilpotents, with descent, and with a cohomology theory adequate to the Weil conjectures. They solved it completely. The techniques were then applied to arithmetic by regarding $\mathrm{Spec}\mathbb{Z}$ as a base scheme like any other — and this is precisely where the theory stops being adequate. $\mathrm{Spec}\mathbb{Z}$ is not a curve. It has no field of constants, no Frobenius, no diagonal in any useful sense, and its "compactification" at the infinite place is not a scheme-theoretic operation at all. The Weil conjectures over a finite field were proved by a trace formula; the analogous statement over $\mathbb{Z}$ has no trace formula because there is no space on which to take a trace.

Every foundational advance of the last two decades can be read as an attempt to supply one missing piece of that structure. Condensed mathematics supplies the analytic completion. Prismatic cohomology supplies the integral p-adic Hodge structure and its Frobenius. The Fargues–Fontaine curve supplies a geometry in which p-adic Galois theory becomes the theory of bundles on a curve. Weil-étale cohomology supplies the correct Euler characteristic. Deninger's programme supplies the dynamical picture. Arakelov theory supplies the archimedean boundary. Λ-rings and $\mathbf{F}_1$-geometry supply the absolute base. Each is a genuine and major advance; none is global; and no two of them are currently glued.

Our proposal is that they are pieces of a single object, and that the gluing is the missing foundation. Whether that is right, the reader will judge. But the proposal is at least of the correct type: it is a proposal about what an arithmetic space *is*, not a proposal about how to compute with the spaces we already have. In that sense, and only in that sense, it aims at the register in which EGA was written.

The Master Identity

The governing idea of the book is one object with two descriptions. To an elliptic curve $E/\mathbb{Q}$ the framework attaches an arithmetic spectrum X_E, and to X_E a determinant line $\Delta(X_E)$: a free module of rank one over $\mathbb{Z}$, built from the cohomology of X_E with its integral lattices. The line carries two trivialisations. The analytic one, λ_{an}, comes from the regularised determinant of the arithmetic flow, and so from the completed L-function: its order of vanishing at $s = 1$ is the multiplicity of an eigenvalue, and its leading coefficient is a determinant. The algebraic one, λ_{alg}, comes from the Selmer complex as the fixed locus of the flow, the Néron–Tate pairing, the periods and the local terms at the bad primes.

The Master Identity, Conjecture A of Chapter 37, asserts that the two agree:

$$\lambda_{\mathrm{an}} \;=\; \lambda_{\mathrm{alg}} \quad \text{in } \Delta_{\mathbb{R}}(X_E),$$

that is, λ_{an} is a $\mathbb{Z}$-basis of $\Delta_{\mathbb{Z}}(X_E)$, sign included. Its hypotheses are printed with it: a solution of the Construction Problem satisfying F1–F12, Conjectures L and M for X_E, and the finiteness of $Ш(E/\mathbb{Q})$, without which the integral line is not defined. From it Chapters 38 and 39 derive the two clauses of the classical conjecture. The rank becomes an index — the dimension of the kernel of the generator of the flow on the critical complex — and the leading coefficient

the determinant of the induced pairing, with the archimedean place on the same footing as the finite ones. The torsion square of the BSD formula appears as Artin–Verdier duality and the Tamagawa product as the vertical local terms; these are two features of the formula that the classical statement leaves unmotivated.

The identity is not easier than the conjecture. The book proves that it is equivalent to the equivariant Tamagawa number conjecture for $h^1(E)(1)$, and Chapter 53 records, as one of its two admissions, that Conjecture A is the BSD formula transcribed. What the reformulation buys is structure: a single integral statement with visible functoriality, and a decomposition of what remains into pieces that can be attacked separately.

Three routes

The programme is pursued along three routes, stated in Chapter 3 in four lines each — what it constructs, what it would give, what it needs, where it stops. They are not three assaults on one wall.

The spectral route (Parts I–III and VI) constructs the FAP substrate, the flow, the categorified trace formula, the spectrum X_E and the fundamental line. It would give the rank formula as an index theorem and the full leading-coefficient formula. It needs the Construction Problem, the consistency of F1–F12, and Conjectures A, G, K, L and M. It stops at Obstruction IV: no action of $\mathbb{R}_{\geq 0}$ with closed orbits of length $\log p$ has ever been constructed on any cohomology over $\mathrm{Spec}\mathbb{Z}$, and the only models of the axioms are $\mathbb{G}_m$ over $\mathbb{Z}$ and curves over finite fields — the two cases where the answer was known beforehand.

The shtuka and Euler-system route (Parts III–IV) constructs a moduli problem of arithmetic shtukas with r legs over a prismatic adelic base, a class $\kappa^{(r)}$ in an exterior bidual of Selmer cohomology, and multi-leg norm relations. It would give the rank lower bound when $L^{(r)}(E,1) \neq 0$ and, with Conjecture F and the theorems of Kato and Skinner–Urban, the p-part of the leading coefficient in the good ordinary case. It needs Conjectures B and F, and it stops at Obstructions IX, X and XII: there is no global Frobenius on $\mathrm{Spec}\mathbb{Z}$, so the isomorphism that closes a shtuka has no known meaning.

The automorphic and theta route (Part V) constructs an r-legged cycle on an arithmetic Shimura-type object, in the tradition of Gross–Zagier, the Kudla programme and the arithmetic fundamental lemma. It would give the archimedean identity relating the self-intersection of the cycle to $L^{(r)}(E,1)/(r!\,\Omega_E)$ — the non-vanishing input without which the second route relates two objects that may both be zero. It needs modularity and a number-field analogue of the theorem of Yun and Zhang, and it stops at Obstruction XI: the function-field proof uses a curve, a Hitchin base and a proper moduli stack, and over $\mathbb{Q}$ there is none of the three.

The relations between the routes are themselves part of the result. The third route is an input to the second, not a parallel to it. Neither the first nor the second implies the other. The spectral route is not logically necessary — Conjectures B, F, H and J would give the numerical formula with no flow at all — so its claim is explanatory: that the rank is an index and that the infinite place becomes a place like the others. And the compatibility of the routes is Conjecture D, rated five stars. Until it is proved, the three routes are three constructions hoped to describe one object.

The architecture of the work

The book is one monograph in ten parts and fifty-six chapters, followed by eight appendices. Part 0 states the problem and the strategy; Parts I–II build the foundations; Parts III–VI apply them to an elliptic curve and derive the classical statements; Parts VII–VIII build the

computational and formal machinery; Part IX states what remains. Each part opens with its plate — a principle, a system and an algorithm — and each chapter closes with a status box stating what it establishes and what it does not.

Part 0, Prologue (*Chapters 1–3*). The conjecture is stated with full precision, the known theorems are inventoried, and Chapter 2 dissects exactly why sixty years of work have produced theorems only in analytic rank at most one: two pipelines, the analytic and the arithmetic, with four points at which they do not communicate. Chapter 3 sets out the strategy — one object with two descriptions — its three routes, the logical spine from the axioms to the classical statements, and the register of twelve conjectures.

Part I, Foundations (*Chapters 4–10*). The category of FAP spaces is constructed from its ingredients: the condensed and analytic substrate, prismatisation and F-gauges, the arithmetic fracture diagram and the adelic gluing that produces an FAP space. Chapter 8 states the twelve axioms F1–F12 — realisation and conservativity, hyperdescent and adelic excision, analytic spectral structure, absolute cyclotomic operations, prismatic gauge completeness, six operations, constructibility and duality, the arithmetic flow, the nuclear trace and regularised determinant, Lefschetz–Verdier localisation, Riemann–Roch orientation and the Selmer fixed-point principle — and says, axiom by axiom, which are theorems in some setting and which are hypotheses. The six-functor formalism, descent and the comparison with the classical realisations follow.

Part II, Higher Structures (*Chapters 11–17*). The categorical machine: motives and the standard conjectures, ∞-categorical arithmetic, dualisability, nuclearity and traces, the Weil-étale fundamental line of Lichtenbaum and Flach–Morin, the arithmetic flow and its infinitesimal generator, the cohomological trace formula, and the regularised determinants from which the completed L-function is to emerge.

Part III, The Arithmetic Spectrum of an Elliptic Curve (*Chapters 18–23*). The theory is applied to one curve. The arithmetic spectrum X_E is specified chart by chart; its cohomology is computed in low degrees; the Selmer group is identified as the fixed locus of the flow; the Néron–Tate height is realised as an arithmetic intersection number on the Arakelov boundary; every local factor of the BSD formula is accounted for; and Chapter 23 recasts the finiteness of Ш as a nuclearity statement.

Part IV, Euler Systems and the Barrier at Rank Two (*Chapters 24–30*). The arithmetic engine. The classical Euler-system machinery and the Iwasawa main conjecture are treated in full; Chapter 25 locates the rank-one barrier; Chapter 27 isolates the rank-two obstruction as a structural statement about the exterior algebra acting on a derived deformation ring. The chapters that follow construct the objects proposed to overcome it — shtukas with legs, arithmetic shtukas over $\mathrm{Spec}\,\mathbb{Z}$, and a higher reciprocity law relating r legs to the r-th derivative of $L(E, s)$.

Part V, The Automorphic and Analytic Side (*Chapters 31–36*). Everything on the L-function side of the identity: modularity and the analytic input it supplies, the Gross–Zagier formula and its descendants, the Kudla programme and arithmetic theta series, relative trace formulae, the higher arithmetic Gross–Zagier problem, and the statistical constraints of the Katz–Sarnak philosophy and of murmurations.

Part VI, The Master Identity (*Chapters 37–41*). The keystone. Conjecture A is stated in full and its conditional consequences, the rank formula and the leading-coefficient formula, are derived from it; the degenerate cases are verified; and Chapter 41 draws the complete logical dependency graph of the book, with an inventory of what has actually been proved.

Part VII, The Computational Architecture (*Chapters 42–47*). A seven-layer system with named modules and explicit interfaces; a certified arithmetic kernel; descent and Selmer engines;

L-function and p-adic engines with algorithms, complexity bounds and error analysis; a discovery layer for machine learning; and a content-addressed certificate format, with a hierarchy of seven levels and a stated trust model.

Part VIII, Formalisation and Machine Reasoning (*Chapters 48–52*). What a formal library can and cannot express about elliptic curves today; autoformalisation and proof search; the multi-agent architecture; the proof-obligation ledger in which every claim of the book is a row; and the trust model of verification.

Part IX, The Programme (*Chapters 53–56*). What remains: the atlas of fourteen obstructions with attack plans; the construction programme in nine stages with explicit success and abandonment criteria; the chapter written against the book, which states what would refute it; and an assessment of prospects that begins from the difficulty ratings.

The eight appendices supply what the chapters should not interrupt themselves to provide: elliptic curves, Galois cohomology and duality, ∞-categories, the condensed and prismatic toolkits, the ledger in full, numerical tables, and the Lean 4 sources. Three indices — of notation, of names and of subjects — close the book.

The register and the atlas

The programme rests on twelve conjectures, A to M without I, each stated formally in its own chapter and rated on a five-point scale, from a good exercise with the framework in hand to a statement that subsumes or exceeds a known major open problem. Eleven of the twelve are rated at the two highest levels. Exactly one, Conjecture H, the integral Hasse principle that assembles prime-by-prime results, is tractable with present technique. Four of the twelve contain a known major open problem under another name: Conjecture A is the equivariant Tamagawa number conjecture for $h^1(E)(1)$ in the notation of the book; Conjecture C is a number-field analogue of the theorem of Yun and Zhang; Conjecture K is the finiteness of Ш; Conjecture L is Deninger's programme in full. A framework that replaces one conjecture by twelve has bought structure at a price, and the book prints the price rather than paraphrasing it.

Fourteen obstructions stand between the framework and the conjecture. Obstructions I–VIII stand in the way of constructing the framework itself: the adelic structured ∞-topos, the coherence of prismatic gauges across all primes, the archimedean place, the arithmetic flow, nuclearity and meromorphic continuation, independence of realisation, the Selmer fixed-point principle, and the equality of the two trivialisations. Obstructions IX–XIV stand on the proposed routes: the existence of arithmetic shtukas over $\mathrm{Spec}\mathbb{Z}$, integral multi-leg norm relations, the higher relative trace identity over $\mathbb{Q}$, primitivity and rank-r descent, wild integral control at small and additive primes, and the mechanism for rank at least two. None is solved. Five contain a known major open problem. Chapter 53 gives each a statement, what it blocks, what it subsumes, its partial results, the strategy that would attack it and its difficulty; Chapter 54 orders the attack into nine stages with gates, so that early stages can fail locally and still leave useful mathematics.

What is proved

The proved content of the book is of three kinds, and the labels keep them apart.

The first is the descent. Assuming the Master Identity and the axioms, Chapter 38 proves the rank formula and Chapter 39 the leading-coefficient formula; the rank lower bound from cycle classes, the equivalence of Conjecture A with the equivariant Tamagawa number conjecture for $h^1(E)(1)$, and the specialisations to ranks 0 and 1 are proved in the same

way. These are complete proofs of conditional statements, labelled [CONDITIONAL], with their hypotheses printed in the statement; every one of them has at least one hypothesis that is unknown for every elliptic curve over $\mathbb{Q}$.

The second is the consistency of the framework with what is known. Chapter 40 audits the twelve axioms over a finite field, axiom by axiom: nine become theorems there, and the formalism specialises to the Grothendieck–Lefschetz trace formula and to the theorem of Tate and Milne; for $\mathbb{G}_m$ over $\mathbb{Z}$ it specialises to the Riemann–Weil explicit formula and the class number formula. The framework is tested, in other words, exactly where the answer was known beforehand. Even there the audit is not uniform: F12 is not a theorem over a finite field, and F8 holds only degenerately.

The third is a body of unconditional results, labelled [NEW] where they are new and each with its proof: the determinant bookkeeping of finite groups and the normalisation lemma relating Λ and L; the isogeny invariance of the algebraic trivialisation, with the product formula as its mechanism; the sign relation and the positivity constraints it imposes; the lemma that the order of vanishing of a regularised determinant at an eigenvalue is its multiplicity; the theorem that Conjecture H is exactly a rationality statement, discharged in analytic rank at most one; the nuclearity of the local transfer operator in the Euler-product region; the rank-two diagnosis of Chapter 27; and the function-field audit itself. Chapter 41 lists them for the sceptical reader, and states what none of them is: a statement about the rank or the leading coefficient of an elliptic curve.

Computation, certificates and formal verification

Parts VII and VIII treat computation and formalisation as part of the mathematics, not as illustration. Every datum of the computational architecture carries a rigour tag, and the tags compose only downward: a certified interval remains certified only if every step that produced it is; a numerical value is never promoted. The certificate hierarchy C1–C7 runs from the bytes of a curve to an audited claim, and its levels measure auditability, not proximity to a proof. The trust hierarchy states what a machine-checked result still assumes after it has been checked.

The formal development fixes the statements, not the proofs. It records which objects a formal library can express today, states the conjecture relative to a continuation datum, and axiomatises the conjectural theory as a typeclass whose consequences can be checked. The Lean 4 sources that accompany the book are published as written; they have not been compiled, and the places where a proof is missing are marked in them. The book does not say that anything in it has been formalised in the sense of a checked proof.

The accompanying Python reference implementation computes the arithmetic of the five running curves of the book — $11.a1$, $37.a1$, $37.b1$, $389.a1$ and $5077.a1$ — with exact arithmetic where the claim is exact, and with certified intervals where it is analytic. Its tests pass, and the numerical tables of Appendix G are the output of that arithmetic, checked independently.

How the programme can fail

A framework that cannot fail tells its reader nothing, and Chapter 55 is written against the book. It sets out the specialisation tests — over a finite field every axiom must be a theorem; in ranks 0 and 1 the constants of Kolyvagin and Gross–Zagier must come out; in the CM case Rubin's results must be reproduced — and the numerical predictions that could fail, of which the Weyl law for the flow is the sharpest single test, followed by the local factors, the Cassels–Tate constraint and the statistics of Selmer groups and murmurations. It states

the structural predictions that a computation could contradict, what a formalisation does and does not certify, and a validation protocol of ordered gates, each with an explicit failure action. A single curve on which the prismatic formula for a local Tamagawa factor disagrees with Tate's algorithm would end that local construction; no appeal to the conjecture could repair the disagreement. Chapter 54 adds the criteria on which a stage of the construction programme is to be abandoned rather than continued.

Chapter 56 then assesses the prospects in the order that the ratings require. It begins from the ratio of the register — eleven of twelve conjectures against, one in favour — separates where the framework is new from where it repackages known mathematics, gives a hostile referee his strongest objections without softening them, and states what the book claims and what it does not.

How this book was made

This book was designed and drafted through a structured collaboration between six independent artificial reasoning systems, each assigned a distinct pillar of the framework and each required to state its results with explicit epistemic tags, followed by a synthesis pass in which the pillars were reconciled, their interfaces made precise, and their conflicts resolved or recorded. The methodology is itself described in Chapter 50, not as an advertisement but because we believe the reader is entitled to know how the text in front of them was produced, and because the multi-agent architecture described there is a component of the proposed research programme rather than merely a means of writing about it.

We note the obvious risk. Systems of this kind produce fluent prose, and fluent prose about mathematics is not mathematics. Our countermeasure is the proof-obligation ledger of Appendix F: every substantive assertion in this book has an entry, a status, a dependency list, and where applicable a formal statement in Lean 4. The reader who wishes to audit the book rather than read it should begin there.

Prerequisites and the path through the book

We assume the theory of elliptic curves at the level of Silverman's two volumes, Galois cohomology at the level of Serre, and algebraic geometry at the level of Hartshorne. Parts I and II require familiarity with ∞-categories at the level of the first two chapters of Lurie's *Higher Topos Theory*; the reader who lacks this will find Appendix C sufficient for the purposes of this book. No prior acquaintance with condensed mathematics, prismatic cohomology, or Weil-étale cohomology is assumed; Appendices D and E develop what is needed. Parts VII and VIII assume no mathematics beyond Parts I–VI, but do assume comfort with reading pseudocode and type signatures.

The dependency chart in the *Leitfaden* shows which chapters may be omitted. Three routes are marked: an *analytic* route for the reader who cares about L-functions, a *foundational* route for the reader who cares about the new geometry, and a *computational* route for the reader who wants to run the engine.

What the Programme takes from this work

For the Grothendieckian Programme this book is a test of method on a problem where the method could not succeed quickly, and its value to the Programme lies in how it records the difficulty. It shows that recursive mathematical discovery, applied without concession, produces three kinds of output that a reader can check: constructions specified exactly enough to be attempted, conditional theorems proved in full, and obstructions stated precisely enough

to be attacked, priced or refuted. It shows that the discipline of status labels, certificates and a proof-obligation ledger can be kept across a work of this length. And it leaves, for the next research subject, the lesson the book states of itself: the measure of AI-assisted mathematics is theorems, and an exact account of where they are missing is part of the mathematics.

Vincent Liu
WiseMotif.ai
2026

Statement of Epistemic Status

The single most important paragraph in this book

The Birch and Swinnerton-Dyer conjecture is *open*. This book does not prove it. This book does not prove the finiteness of the Tate–Shafarevich group. This book does not prove the rank part of the conjecture in any case not already known. No theorem in this book, read correctly, asserts otherwise. A reader who comes away with the impression that a Millennium Problem has been solved has misread the book, and the fault will be ours.

I. The four tags

Mathematical writing ordinarily leaves the epistemic status of an assertion to be inferred from context: a numbered Theorem is proved, a numbered Conjecture is not, and everything else is a matter of the reader's judgement. In a work of this kind, which proposes new foundations and derives consequences within them, that convention is inadequate. A statement may be a theorem *of the new framework* while the framework itself is conjectural; a statement may be a classical theorem imported without change; a statement may hold under a named standard conjecture. These are entirely different things and the reader must be able to tell them apart at a glance.

We therefore tag every substantive assertion. The four tags are:

[KNOWN]
: The assertion is a theorem of the existing literature, proved unconditionally under the hypotheses explicitly stated. A citation is always given. If the hypotheses of the cited theorem are stronger than ours, we say so.

[CONDITIONAL]
: The assertion is a theorem *conditional on a named conjecture*, or is itself a standard conjecture of the field. The conjecture on which it depends is always named — "conditional on the finiteness of Ш", "conditional on the Iwasawa Main Conjecture at p", "conditional on the standard conjectures" — and never left implicit. A conditional theorem whose hypothesis is one of *our own* new conjectures is tagged **[CONDITIONAL]** and the hypothesis is cross-referenced to its **[NEW]** statement.

[NEW]
: The assertion is a definition, axiom, construction, or conjecture introduced in this book. A **[NEW]** tag is *never* a claim of truth. It is a claim of novelty and of precision: that the statement is well-posed, that its ingredients are defined, and that it has not, to our knowledge, been made before in this form. Where a **[NEW]** statement is a variant or strengthening of an existing conjecture, the antecedent is identified.

[OPEN]
: The assertion is a problem we pose and do not solve. Chapter 53 is an atlas of these.

A tag applies to the whole of the statement it decorates. Where a theorem has several parts of different status, the parts are tagged separately.

II. What the framework does and does not deliver

It is worth setting out, in one place and without hedging, the logical position of the book's central object.

The *Master Identity* (Conjecture A, Chapter 37) is a conjecture. It is stated in terms of an object X_E, the FAP arithmetic spectrum of E, whose existence is the content of the *Construction Problem* (Problem 7.6.1). The Construction Problem is [OPEN]. We reduce it to eight named obstructions (Chapter 53), of which:

- three (I, II, VI) we regard as hard but tractable with present technology, and for each we give a concrete programme with named lemmas;
- three (III, V, VIII) we regard as difficult, requiring new ideas of a kind we can describe but not supply;
- two (IV, VII) subsume, respectively, the substance of Deninger's programme and the finiteness of Ш, and are therefore *at least as hard as* major open problems in their own right.

Consequently the implication

$$\text{Construction Problem solved} \;+\; \text{Conjecture A} \;\Longrightarrow\; \text{BSD}$$

is, we believe, correct and is proved in Chapters 38–39 [CONDITIONAL]. The implication is not a proof of BSD, and its value lies entirely in whether the left-hand side is a better target than the right. We argue in Chapter 3 that it is, and we make the argument falsifiable in Chapter 55.

III. On the reading of "more revolutionary than EGA"

The framework proposed here is more *radical* than EGA in the specific sense that it revises the notion of arithmetic space itself, rather than building theory on top of a fixed notion. It is not more *accomplished* than EGA, and the comparison should not be read as a comparison of achievement. EGA is a completed edifice whose theorems are proved. This book is a design document with an implementation, a list of unproved statements, and an argument that the list is the right one.

The reader who wants a single sentence may take this one: *we have tried to change the question, and we have tried to be exact about what changing the question does and does not accomplish.*

IV. Provenance and audit

The drafting methodology is described in the Preface, §V, and analysed in Chapter 50. Because the text was produced with substantial machine assistance, we have adopted three countermeasures which the reader should be aware of and should use:

(i) **The proof-obligation ledger** (Appendix F). Every substantive assertion has an identifier PO-NNNN, a status, a dependency list, a citation where applicable, and a formalisation state. The ledger is machine readable and ships with the source distribution.

(ii) **The citation audit.** Claims about the existing literature are the ones most vulnerable to confabulation. Every citation of a specific theorem carries author, year, and the name of the result. Claims about the current contents of a formal library (`mathlib`) are tagged `[STATE]` and dated, because they decay; Appendix H includes a script that re-audits them.

(iii) **The adversarial pass.** Chapter 55 is written against the book: it lists the ways in which the framework could be wrong, the numerical experiments that would show it, and the

structural facts that would refute it outright. A framework that cannot be refuted is not offered here as a contribution.

V. Notation for status inside statements

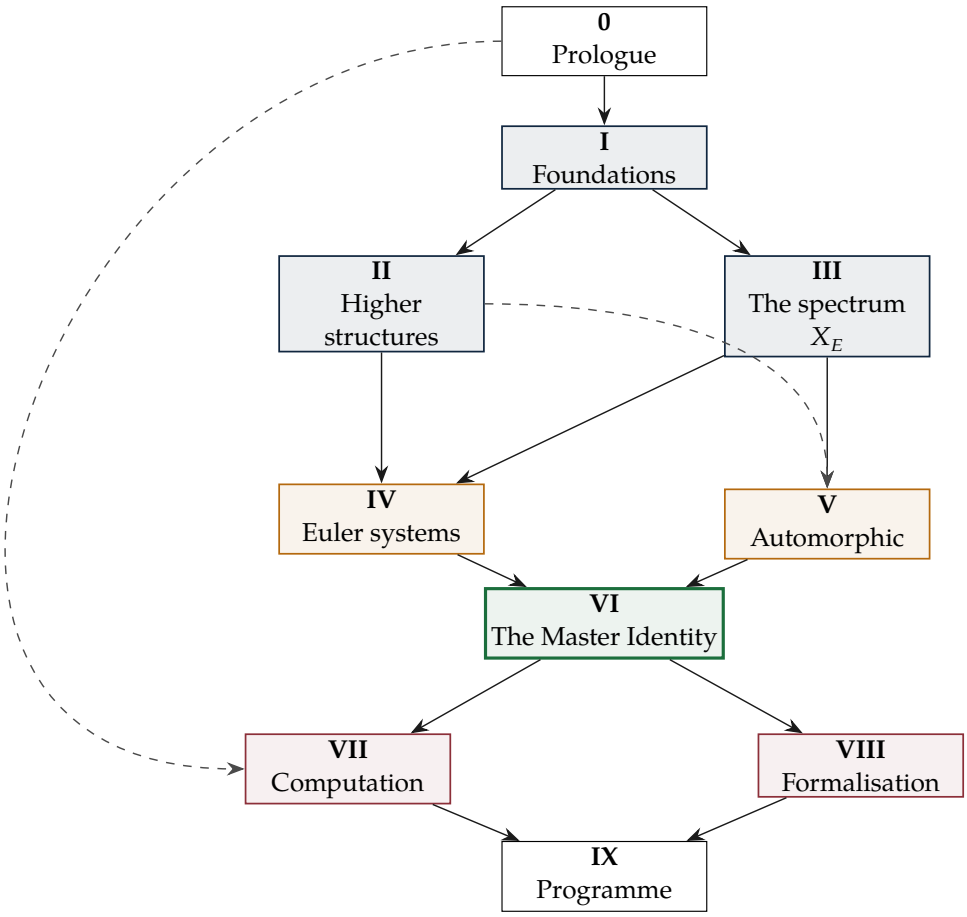
Within the text, tags appear immediately after the theorem number:

> **Theorem 25.3.1** [KNOWN] (*Kolyvagin 1988*). Let $E/\mathbb{Q}$ be a modular elliptic curve and suppose $\mathrm{ord}_{s=1} L(E,s) \le 1$. Then $\mathrm{rank}\, E(\mathbb{Q}) = \mathrm{ord}_{s=1} L(E,s)$ and $Ш(E/\mathbb{Q})$ is finite.
>
> **Conjecture** 30.2.1 [NEW] (*Higher Reciprocity Law*). …
>
> **Theorem 39.1.2** [CONDITIONAL] (*conditional on Conjecture A and Problem 7.6.1*). …

Where a chapter is entirely expository of known material, this is stated in its opening abstract and individual tags are omitted.

Leitfaden

This book is long, and few readers will want all of it. The chart below records the logical dependencies between its ten parts; the routes that follow describe four self-contained paths through the material.



Solid arrows are hard dependencies; dashed arrows are optional enrichments. Part VII may be read directly after the Prologue by a reader interested only in the computational engine.

Four routes

The foundational route.
Chapters 1–3, 4–10, 11–17, 37, 53–54. This is the route for the reader who wants the new geometry: FAP spaces, the twelve axioms, the six-functor formalism, the arithmetic flow, the categorified trace formula, and the statement of the Master Identity, followed by an account of what remains unbuilt. Roughly 190 pages.

The arithmetic route.
Chapters 1–3, 18–23, 24–30, 38, 41. This is the route for the reader who works in Iwasawa theory or on Euler systems: the arithmetic spectrum and its cohomology, Selmer groups as a fixed locus, the classical Euler system machinery, the precise diagnosis of the rank-two obstruction, arithmetic shtukas over $\mathrm{Spec}\mathbb{Z}$, and the Higher Reciprocity Law. Roughly 180 pages. Parts I and II may be consulted rather than read; Chapter 18 recalls everything from them that is used.

The analytic route.
Chapters 1–3, 31–36, 32–35, 39–40. For the reader interested in L-functions, Gross–Zagier formulae, arithmetic theta series, relative trace formulae, and the statistics of ranks. Roughly 120 pages. This route can be read with no knowledge of Parts I and II beyond the summary in Chapter 3.

The computational route.
Chapters 1–3, 42–47, 48–52, 55. For the reader who wants to build or run the engine: the seven-layer architecture, the certified arithmetic kernel, the descent, Selmer, L-function and p-adic engines, the discovery layer, the certificate format, the Lean 4 development, and the validation protocol. Roughly 150 pages, and essentially independent of Parts I–VI.

Chapters that may be omitted on a first reading

Chapters 9, 12, 13, 22, 26, 34, 43, 45, 49 and 52 are technical and are used later only through their statements. Each of them opens with a summary box listing exactly which of its results are needed downstream; a reader in a hurry may read the box and move on.

Where the new material is

The reader who wants only what is new should read Chapter 7, Chapter 8, Chapter 15, Chapter 16, Chapter 27, Chapter 29, Chapter 30, Chapter 35, Chapter 37 and Chapter 53. That is about 100 pages and contains every `[NEW]`-tagged statement of consequence in the book. Everything else is either standard material presented in the language of the framework, or the machinery required to make those ten chapters precise.

Notation and Conventions

I. General conventions

Rings.
All rings are commutative with unit unless the contrary is said. "Ring" in a derived or spectral context means E_∞-ring; the distinction is always made explicit where it matters.

Categories.
Ordinary categories are written $\mathcal{C}, \mathcal{D}, \dots$; ∞-categories are written $\mathbf{C}, \mathbf{D}, \dots$. We work in the quasicategorical model throughout, following Lurie, and we suppress coherence data in the notation without further comment. "Colimit" means homotopy colimit; colim is reserved for it.

Complexes.
Cohomological indexing throughout. $\mathrm{R}\Gamma$ denotes the derived global sections functor and $H^i(-) = H^i(\mathrm{R}\Gamma(-))$.

Sheaves.
$\mathcal{O}$ is a structure sheaf; a superscript records the flavour, thus $\mathcal{O}^{\mathrm{sp}}$ for the spectral structure sheaf and $\mathcal{O}^{\blacksquare}$ for the solid one.

Primes.
The letter p always denotes a prime number; ℓ denotes a prime, usually distinct from p or from the residue characteristic. The letter v denotes a place of a number field, including the archimedean ones; $v \nmid \infty$ and $v \mid \infty$ distinguish them.

Epistemic tags.
[KNOWN], [CONDITIONAL], [NEW], [OPEN], as defined in the *Statement of Epistemic Status*. Every substantive assertion carries one.

II. Classical arithmetic objects

Symbol	Meaning
E, $E/\mathbb{Q}$	an elliptic curve over $\mathbb{Q}$, given by a global minimal Weierstrass model
$N = N_E$	the conductor of E
Δ_E	the minimal discriminant
$E(\mathbb{Q})$	the Mordell–Weil group
$r_{\mathrm{alg}} = \operatorname{rank} E(\mathbb{Q})$	the algebraic rank
$r_{\mathrm{an}} = \operatorname{ord}_{s=1} L(E,s)$	the analytic rank
$E(\mathbb{Q})_{\mathrm{tors}}$	the torsion subgroup
$\hat{h}$, $\langle\,,\,\rangle_{\mathrm{NT}}$	the canonical Néron–Tate height and its associated pairing
$\operatorname{Reg}(E/\mathbb{Q})$	the regulator, $\det\langle P_i, P_j\rangle_{\mathrm{NT}}$ computed on a basis of $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}$
Ω_E	the real period $\int_{E(\mathbb{R})} \lvert\omega\rvert$ for the Néron differential ω
c_p	the Tamagawa number at p, $[E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)]$
$Ш(E/\mathbb{Q})$	the Tate–Shafarevich group
$\mathrm{Sel}_n(E/\mathbb{Q})$, Sel_{p^∞}	the n-Selmer and p^∞-Selmer groups
a_p	the trace of Frobenius, $a_p = p + 1 - \#\widetilde{E}(\mathbb{F}_p)$ for $p \nmid N$
$L(E,s)$	the Hasse–Weil L-function
$\Lambda(E,s)$	the completed L-function $N^{s/2}(2\pi)^{-s}\Gamma(s)L(E,s)$
$\varepsilon(E)$	the sign of the functional equation
T_pE, V_pE	the p-adic Tate module, and $V_pE = T_pE \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$
$H^1_f(\mathbb{Q}, V)$	the Bloch–Kato Selmer group

III. Objects introduced in this book

Every symbol in this table is **[NEW]**. The chapter of definition is given, and the canonical label is recorded so that the reader may locate the definition by cross-reference rather than by page.

Symbol	Meaning	Defined
$\mathfrak{X}$	a Frobenius–Arakelov–Prismatic (FAP) space	Ch. 7
FAP	the ∞-category of FAP spaces	Ch. 7
$\mathcal{J}_{\mathrm{ar}}$	the arithmetic fracture diagram: the indexing ∞-category with vertices η, ∞, p^{Δ}, p^{HT}, p^{FF}, p^{an}	Ch. 7
$\overline{S}$	the Arakelov compactification of $\operatorname{Spec}\mathbb{Z}$	Ch. 4
X_E	the BSD arithmetic spectrum of E	Ch. 18
$\mathrm{R}\Gamma_{\mathrm{FAP}}(X_E, -)$	FAP cohomology	Ch. 19
Φ^t	the arithmetic flow	Ch. 15
Θ	its infinitesimal generator, $\Theta = \frac{d}{dt}\Phi^t\big\vert_{t=0}$	Ch. 15
ψ^n	the absolute cyclotomic (Λ-)operations; ψ^p induces Frobenius at p	Ch. 8
$\det_\infty$	the regularised determinant on nuclear objects	Ch. 17
$\Delta(X_E)$	the FAP fundamental line	Ch. 37
λ_{an}, λ_{alg}	the analytic and algebraic trivialisations of $\Delta(X_E)$	Ch. 37
$\mathrm{Sht}^{(r)}_{\mathbb{Z}}$	the stack of r-legged arithmetic shtukas over $\operatorname{Spec}\mathbb{Z}$	Ch. 29
$\kappa^{(r)}$	the r-legged determinant class of the shtuka system	Ch. 29
$\langle\,,\,\rangle^{\mathrm{der}}$	the derived height pairing	Ch. 30
PO-NNNN	a proof-obligation identifier	App. F

IV. Imported notation

Symbol	Meaning
$\mathrm{Cond}(C)$, $\blacksquare$	condensed objects; the solid tensor structure (Clausen–Scholze)
Nuc	nuclear objects in a condensed or analytic setting
(A, I), $\mathrm{R}\Gamma_{\Delta}(X/A)$	a prism; prismatic cohomology (Bhatt–Scholze 2022)
X^{Δ}, X^{HT}	the prismatisation and the Hodge–Tate stack (Bhatt–Lurie; Drinfeld)
$\mathcal{N}^{\bullet}$	the Nygaard filtration
φ	Frobenius, in whichever theory is under discussion
$X^{\diamond}$	the diamond attached to an adic space (Scholze)
X_{FF}	the Fargues–Fontaine curve
$\mathrm{R}\Gamma_{W}(\overline{X}, -)$	Weil-étale cohomology (Lichtenbaum; Flach–Morin)
THH, TC, TP	topological Hochschild homology; topological cyclic homology; periodic topological cyclic homology
$\mathbf{SH}(S)$, $\mathbf{DM}(S)$	the motivic stable homotopy category; Voevodsky's derived category of motives
$\mathrm{CH}^{i}(X)$, $\widehat{\mathrm{CH}}^{i}(X)$	the Chow group; the arithmetic Chow group (Gillet–Soulé)
$\mathrm{char}(M)$	the characteristic ideal of a torsion Iwasawa module
$\Lambda_{\mathrm{Iw}} = \mathbb{Z}_{p}[[\Gamma]]$	the Iwasawa algebra

V. Numbering

Definitions, theorems, propositions, lemmas, corollaries, conjectures, axioms, remarks, examples and problems share a single counter, numbered *chapter.section.item*. Thus the item numbered 7.3.1 is the first numbered item of Section 7.3 and the item numbered 7.3.2 the second. Equations are numbered by section; only equations that are referred to are numbered. Figures and tables are numbered by chapter.

Cross-references are hyperlinked in the electronic version. A reference of the form PO-0142 points to an entry of the proof-obligation ledger in Appendix F, where its status, dependencies and formalisation state are recorded.

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Part 0

Prologue

The Problem, the Obstruction, and the Programme

Three chapters that state the Birch and Swinnerton-Dyer conjecture with full precision, dissect exactly why sixty years of work has produced theorems only in analytic rank at most one, and set out the strategy of the book: to replace the comparison of two unrelated quantities by the construction of a single object of which both are realisations.

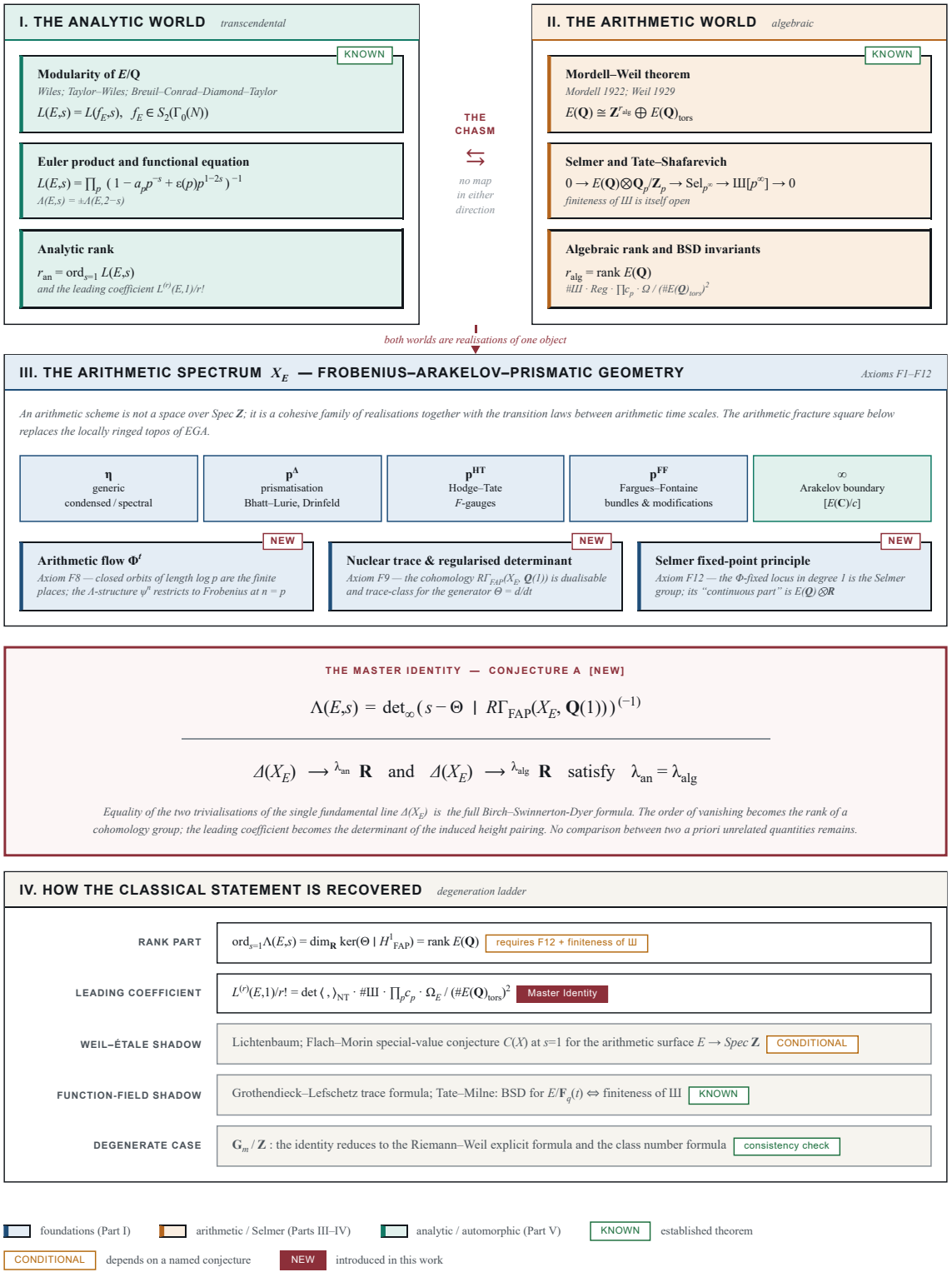


Figure 0.1. The Spectral Descent Principle: BSD as a single determinant identity.

What it does not show: a proof — Conjecture A is not proved here, and its remaining content is tracked in the proof-obligation ledger of Appendix F.

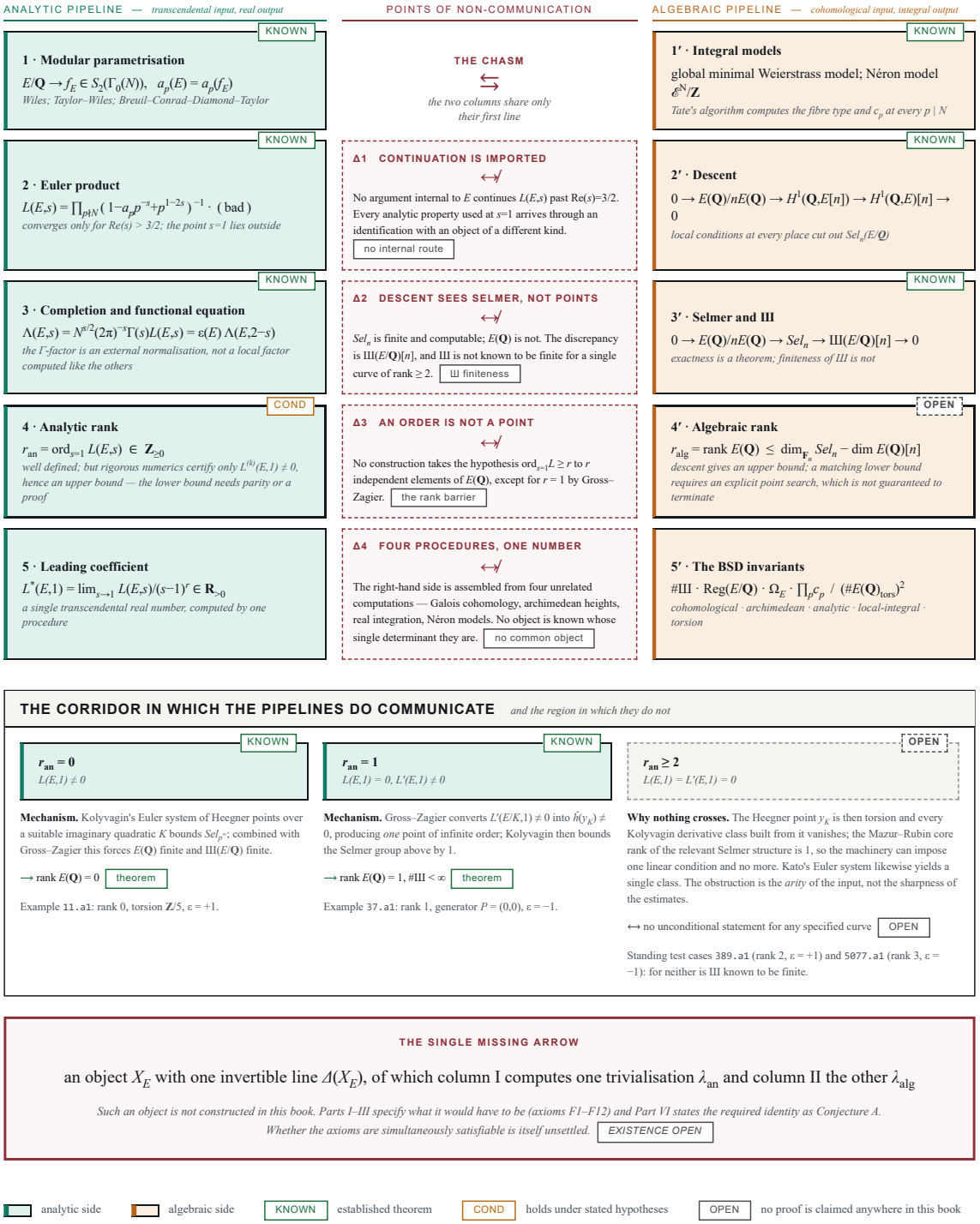
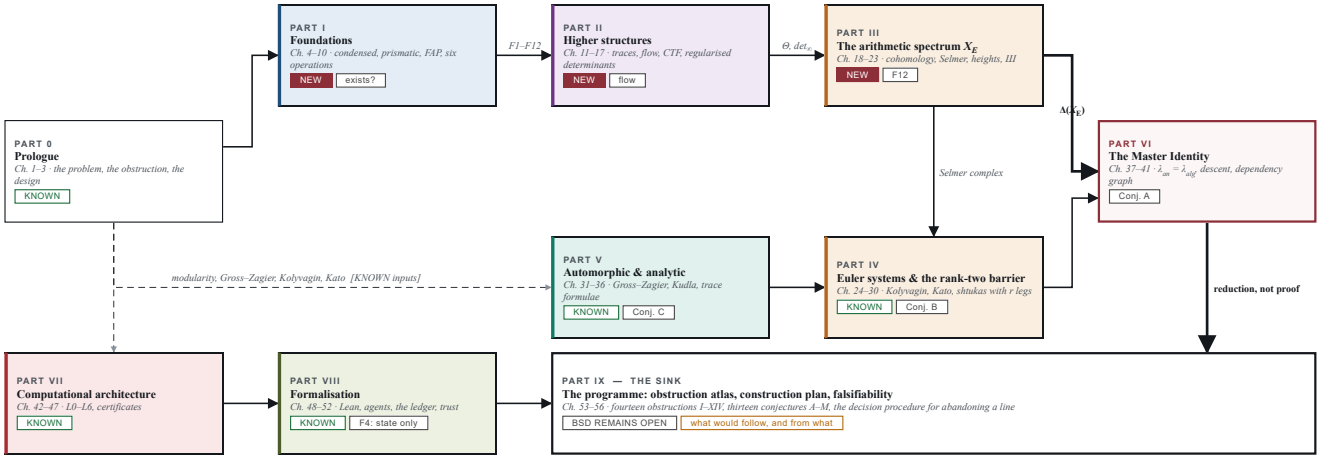


Figure 0.2. Anatomy of the obstruction: two pipelines, four points of non-communication, one missing arrow.

What it does not show: lemmas awaiting proof — each of the four central cells records the absence of a map between categories of objects.



WHAT EACH PART MUST DELIVER, AND WHAT IT MUST CLEAR FIRST *obstruction numerals are those of the atlas, Chapter 53*

PART	CHAPTERS	DELIVERABLE	FEEDS	OBSTRUCTIONS IT MUST CLEAR	STATUS
0 Prologue	1–3	The inventory of what is known; the diagnosis that the defect is foundational; the design brief.	—	— (expository)	KNOWN
I Foundations	4–10	The fracture diagram $\mathscr{A}_{ar}$, the category **FAP**, the topology τ_{FVA}, and axioms F1–F12 as a specification.	A	**I** adelic ∞-topos; **II** gauge coherence over all p; **III** the archimedean fibre; **VI** realisation independence	NEW satisfiable?
II Higher structures	11–17	Dualisability and nuclearity; the flow Φ^t and generator Θ; the categorified trace formula; $\det_\infty$ and $\mathbb{Z}_E(s)$.	A	**IV** construction of the flow (subsumes Deninger's programme); **V** nuclearity and meromorphic continuation	NEW IV, V
III Arithmetic spectrum	18–23	X_E, its cohomology, the critical complex $\mathcal{C}_E$, Selmer as a fixed locus, the Θ-Bockstein height, local terms.	A	**VII** the Selmer fixed-point principle F12 — subsumes finiteness of Ш and ETNC	NEW VII
IV Euler systems	24–30	The rank-two diagnosis (a theorem); the stack $\mathrm{Sht}^{(r)}_{\mathbf{Z}}$; the class $\kappa^{(r)}$; Conjecture B.	B	**IX** existence of $\mathrm{Sht}^{(r)}_{\mathbf{Z}}$; **X** integral multi-leg norm relations; **XII** rank-r descent; **XIV** the mechanism for $r \geq 2$	KNOWN IX, X, XII, XIV
V Automorphic	31–36	Modularity; Gross–Zagier and its descendants; arithmetic theta; relative trace formulae; Conjecture C.	C	**XI** a higher relative trace identity over **Q** — a number-field analogue of Yun–Zhang	KNOWN XI
VI Master Identity	37–41	$\Delta(X_E)$ and its two trivialisations; the descent to the rank and leading-coefficient statements; the dependency graph itself.	A + B + C	**VIII** equality of the two trivialisations (ETNC-strength); **XIII** wild integral control at $p = 2, 3$ and additive primes	OPEN
VII Computation	42–47	Certified arithmetic kernel; descent, Selmer and L-function engines; the rigour-tag algebra; certificates C1–C7.	evidence	practical only: **P1**–**P5**. The engine can verify shadows of the framework, never the framework itself.	KNOWN
VIII Formalisation	48–52	The ladder F0–F4; the proof-obligation ledger; the trust model TCB-0–3; the multi-agent architecture.	fidelity	**P6**: Néron models, heights, Selmer, Ш and modularity are all absent from mathlib. F4 can *state* the conjectures, never prove them.	KNOWN
IX Programme	53–56	The obstruction atlas I–XIV; the ordered construction plan; the falsification experiments; the honest prospects.	—	All fourteen remain open. Ten of the thirteen conjectures are graded ★★★★ or worse.	OPEN

KNOWN theorem of the existing literature · COND true under explicitly stated hypotheses · NEW constructed and proved in this book · OPEN conjecture or obstruction; no proof claimed

Figure 0.3. Leitfaden: the programme in ten parts, with its conjectural load and its obstructions.

What it does not show: an edge that ends at BSD — Part VI delivers a reduction, valid only if a FAP space satisfying F1–F12 exists, Conjectures A, B and C hold, and Ш$(E/\mathbb{Q})$ is finite.

Chapter 1

The Conjecture of Birch and Swinnerton-Dyer

We state the Birch and Swinnerton-Dyer conjecture in the form in which it will be used throughout this book, assemble the classical apparatus on which the statement depends, and record with precision what is known. The chapter is expository: every result in it is a theorem of the existing literature, and individual epistemic tags are therefore used only where the reader might otherwise mistake a conditional statement for an unconditional one. Its purpose is to fix notation, to make explicit the hypotheses hidden inside the customary abbreviations, and to establish the inventory of known results against which the framework of this book must be measured.

1.1 The empirical origin

Between 1958 and 1965, Bryan Birch and Peter Swinnerton-Dyer used the EDSAC II computer at the University of Cambridge Mathematical Laboratory to tabulate, for a large family of elliptic curves over $\mathbb{Q}$, the quantity

$$\pi_E(X) \;=\; \prod_{p \le X,\; p \nmid N} \frac{N_p}{p}, \qquad N_p \;=\; \#\widetilde{E}(\mathbb{F}_p), \tag{1.1.1}$$

where $\widetilde{E}$ denotes the reduction of a global minimal model of E modulo p. Their observation [BSD65] was that

$$\pi_E(X) \;\sim\; C_E\,(\log X)^{r} \qquad (X \to \infty)$$

for a constant $C_E > 0$ and an integer r which, in every case they could test, equalled the rank of the finitely generated abelian group $E(\mathbb{Q})$.

The heuristic behind (1.1.1) is transparent and worth recording, because it explains why the conjecture has the shape it has. If $E(\mathbb{Q})$ has large rank then E has many rational points; rational points reduce to points modulo p for almost all p; so one expects N_p to be larger than average, and the product $\prod N_p/p$ to diverge the faster the larger r is. The passage from this heuristic to a precise statement requires the analytic continuation of the Hasse–Weil L-function, which in 1965 was not available for a single curve without complex multiplication. The conjecture was therefore, at its birth, a statement about an object not known to exist.

Remark 1.1.1. The heuristic is, taken literally, false: the convergence in (1.1.1) is extremely delicate, and Goldfeld [Gol82] showed that the asymptotic $\pi_E(X) \sim C_E(\log X)^r$, with a suitable interpretation of the constant, is equivalent to the Riemann hypothesis for $L(E,s)$ together with the rank part of the conjecture. The naive product is thus not a shortcut; it encodes more than the conjecture itself. This is the first of many occasions in this book on which an apparently elementary reformulation turns out to be strictly harder than the statement it was meant to simplify.

1.2 The arithmetic side

1.2.1 The Mordell–Weil group

Let E be an elliptic curve over $\mathbb{Q}$, which for us always means a smooth projective geometrically connected curve of genus one over $\mathbb{Q}$ equipped with a rational point O, together with the group law for which O is the identity. We fix throughout a global minimal Weierstrass model

$$y^2 + a_1xy + a_3y \;=\; x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in \mathbb{Z}, \tag{1.2.1}$$

with discriminant Δ_E of minimal absolute value, and we write ω_E for the associated Néron differential $\omega_E = dx/(2y + a_1x + a_3)$.

Theorem 1.2.1 (Mordell; Weil). *[KNOWN] The group $E(\mathbb{Q})$ is finitely generated. More generally, for any number field K and any abelian variety A/K, the group $A(K)$ is finitely generated.*

The case $E/\mathbb{Q}$ is Mordell's theorem of 1922 [Mor22]; the general case is Weil's thesis of 1929 [Wei29]. We write

$$E(\mathbb{Q}) \;\cong\; \mathbb{Z}^{\,r_{\mathrm{alg}}} \oplus E(\mathbb{Q})_{\mathrm{tors}}, \qquad r_{\mathrm{alg}} \;=\; \operatorname{rank} E(\mathbb{Q}), \tag{1.2.2}$$

and call r_{alg} the *algebraic rank*.

Two features of Theorem 1.2.1 deserve emphasis, because they recur at every level of this book. First, the proof is a *descent*: one shows that $E(\mathbb{Q})/nE(\mathbb{Q})$ is finite by embedding it in a Galois cohomology group, and then deduces finite generation from a height argument. Second, the proof is *ineffective* for the rank: it produces a bound on the size of a generating set only after one knows that a certain Galois cohomology group, the Tate–Shafarevich group, is finite — and that finiteness is not known. The gap between what descent computes and what one wants to compute is precisely the Tate–Shafarevich group, and it is the same gap that separates the two sides of the conjecture.

Theorem 1.2.2 (Mazur). *[KNOWN] The torsion subgroup $E(\mathbb{Q})_{\mathrm{tors}}$ is isomorphic to one of the fifteen groups*

$$\mathbb{Z}/n\mathbb{Z} \quad (1 \le n \le 10,\ n = 12), \qquad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2m\mathbb{Z} \quad (1 \le m \le 4),$$

and each occurs for infinitely many $\mathbb{Q}$-isomorphism classes of elliptic curves.

This is Mazur's theorem [Maz77; Maz78]. By contrast, no bound whatever is known for r_{alg}: it is not known whether the rank of an elliptic curve over $\mathbb{Q}$ is bounded, and the largest exhibited rank at the time of writing is 29, due to Elkies and Klagsbrun. The contrast between Theorem 1.2.2 and this state of ignorance is instructive. Torsion is a *local* phenomenon, controlled by the geometry of modular curves; rank is a *global* phenomenon, and no geometric object is known whose points control it.

1.2.2 Selmer and Tate–Shafarevich groups

Let $n \ge 2$. The Kummer sequence for multiplication by n on E over $\overline{\mathbb{Q}}$ yields, after taking $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-cohomology, a short exact sequence

$$0 \longrightarrow E(\mathbb{Q})/nE(\mathbb{Q}) \longrightarrow H^1(\mathbb{Q}, E[n]) \longrightarrow H^1(\mathbb{Q}, E)[n] \longrightarrow 0. \tag{1.2.3}$$

Imposing the local conditions coming from the corresponding sequences over each completion $\mathbb{Q}_v$ gives the *n-Selmer group*

$$\mathrm{Sel}_n(E/\mathbb{Q}) \;=\; \ker\Bigl(H^1(\mathbb{Q}, E[n]) \longrightarrow \prod_{v} H^1(\mathbb{Q}_v, E)\Bigr), \tag{1.2.4}$$

and the *Tate–Shafarevich group*

$$Ш(E/\mathbb{Q}) \;=\; \ker\Big(H^1(\mathbb{Q},E) \longrightarrow \prod_v H^1(\mathbb{Q}_v,E)\Big). \tag{1.2.5}$$

These fit into the fundamental exact sequence

$$0 \longrightarrow E(\mathbb{Q})/nE(\mathbb{Q}) \longrightarrow \mathrm{Sel}_n(E/\mathbb{Q}) \longrightarrow Ш(E/\mathbb{Q})[n] \longrightarrow 0. \tag{1.2.6}$$

Theorem 1.2.3. *[KNOWN]* $\mathrm{Sel}_n(E/\mathbb{Q})$ *is finite for every* $n \geq 2$*, and is effectively computable.*

The finiteness is classical (Weil); effectivity for $n = 2$ goes back to Birch–Swinnerton-Dyer and Cremona [Cre97], and explicit n-descent for $n = 3, 4, 8, 12$ is due to Cremona, Fisher, O'Neil, Simon, Stoll and Watkins [CFO+08]. The point of (1.2.6) is that the computable object Sel_n bounds the desired object $E(\mathbb{Q})/nE(\mathbb{Q})$ from above, with error measured exactly by $Ш[n]$.

Conjecture 1.2.4 (Shafarevich–Tate). *[OPEN]* $Ш(E/\mathbb{Q})$ *is finite.*

Warning 1.2.5. Theorem 1.2.4 is open in general. It is known when $\mathrm{ord}_{s=1}L(E,s) \leq 1$, by the combination of Gross–Zagier and Kolyvagin (Theorem 1.5.3 below), and in no case of rank ≥ 2. In particular the statement "Ш is finite" may never be used in this book without either a hypothesis on the analytic rank or an explicit [CONDITIONAL] tag. The reader is warned that a great deal of the literature states results "assuming Ш finite" without repeating the hypothesis in the theorem statement; we do not follow that convention.

1.2.3 Heights and the regulator

The canonical height is the function

$$\hat{h} : E(\mathbb{Q}) \longrightarrow \mathbb{R}_{\geq 0}, \qquad \hat{h}(P) \;=\; \lim_{n\to\infty} \frac{h_x(2^n P)}{4^n}, \tag{1.2.7}$$

where h_x is the naive logarithmic height of the x-coordinate. It is a positive definite quadratic form on $E(\mathbb{Q}) \otimes \mathbb{R}$ vanishing exactly on torsion, and it decomposes as a sum of local Néron heights $\hat{h} = \sum_v \lambda_v$. Setting $\langle P, Q\rangle_{\mathrm{NT}} = \frac{1}{2}(\hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q))$, the *regulator* is

$$\mathrm{Reg}(E/\mathbb{Q}) \;=\; \det\big(\langle P_i, P_j\rangle_{\mathrm{NT}}\big)_{1\leq i,j\leq r_{\mathrm{alg}}}, \tag{1.2.8}$$

computed on any basis $P_1, \dots, P_{r_{\mathrm{alg}}}$ of $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}$; it is independent of the choice, and equals 1 by convention when $r_{\mathrm{alg}} = 0$.

That $\hat{h}$ is a sum of local terms, one for each place of $\mathbb{Q}$ including the archimedean one, is the first appearance in this book of the phenomenon that drives all of Part III: the arithmetic invariants of E are already adelic, and the archimedean place is not an afterthought but a constituent. In Chapter 21 we shall see $\hat{h}$ as an arithmetic intersection number in the sense of Arakelov theory, and $\mathrm{Reg}(E/\mathbb{Q})$ as the covolume of a lattice in a real vector space attached to the Arakelov boundary of the arithmetic spectrum.

1.2.4 Local invariants

For each prime p let $E_0(\mathbb{Q}_p) \subseteq E(\mathbb{Q}_p)$ denote the subgroup of points with non-singular reduction. The *Tamagawa number* at p is

$$c_p \;=\; [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)], \tag{1.2.9}$$

a positive integer equal to 1 for all p of good reduction, and computable in all cases by Tate's algorithm [Tat75]. The *real period* is

$$\Omega_E \;=\; \int_{E(\mathbb{R})} |\omega_E|, \tag{1.2.10}$$

which for a minimal model is the (possibly doubled, according to the number of real components) fundamental real period of the Néron lattice, computable to arbitrary precision by the arithmetic–geometric mean.

1.3 The analytic side

1.3.1 The Hasse–Weil L-function

For $p \nmid N$ set $a_p = p + 1 - \#\widetilde{E}(\mathbb{F}_p)$, and for $p \mid N$ set $a_p = 1, -1, 0$ according as E has split multiplicative, non-split multiplicative, or additive reduction at p. The Hasse–Weil L-function is

$$L(E,s) \;=\; \prod_{p \mid N} (1 - a_p p^{-s})^{-1} \prod_{p \nmid N} (1 - a_p p^{-s} + p^{1-2s})^{-1}. \tag{1.3.1}$$

By Hasse's theorem $|a_p| \le 2\sqrt{p}$, so the Euler product converges absolutely for $\Re(s) > 3/2$. The point $s = 1$ at which the conjecture is stated lies strictly outside the region of convergence; the entire content of the analytic side depends on continuing $L(E,s)$ past it.

Theorem 1.3.1 (Wiles; Taylor–Wiles; Breuil–Conrad–Diamond–Taylor). *[KNOWN] Every elliptic curve $E/\mathbb{Q}$ of conductor N is modular: there is a newform $f_E \in S_2(\Gamma_0(N))$ with $L(E,s) = L(f_E,s)$. Consequently $L(E,s)$ extends to an entire function on $\mathbb{C}$ and the completed L-function*

$$\Lambda(E,s) \;=\; N^{s/2} (2\pi)^{-s} \Gamma(s) L(E,s) \tag{1.3.2}$$

satisfies $\Lambda(E,s) = \varepsilon(E)\,\Lambda(E,2-s)$ with $\varepsilon(E) \in \{\pm 1\}$.

The semistable case is Wiles [Wil95] together with Taylor–Wiles [TW95]; the general case is Breuil–Conrad–Diamond–Taylor [BCDT01]. We may therefore define the *analytic rank*

$$r_{\mathrm{an}} \;=\; \operatorname{ord}_{s=1} L(E,s), \tag{1.3.3}$$

a well-defined non-negative integer.

Remark 1.3.2. It is worth pausing on what Theorem 1.3.1 costs. The analytic continuation of $L(E,s)$ is not obtained by any argument internal to the theory of E; it is obtained by identifying $L(E,s)$ with the L-function of a modular form, an object of an entirely different kind, via a deformation-theoretic argument about Galois representations. Every analytic property of $L(E,s)$ used in this book — continuation, functional equation, growth in vertical strips, the very meaningfulness of $s = 1$ — is imported through this identification. The framework of Part I is designed so that these properties become *internal*: in Chapter 17 the continuation of $L(E,s)$ becomes a statement about the spectrum of an operator, and the functional equation becomes a duality. Whether that design succeeds is the subject of the obstruction analysis in Chapter 56.

1.3.2 Parity

Theorem 1.3.3 (Nekovář; Dokchitser–Dokchitser). *[KNOWN] Let $E/\mathbb{Q}$ and let p be a prime. Then*

$$(-1)^{\operatorname{rank}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(E/\mathbb{Q})} \;=\; \varepsilon(E),$$

where $\operatorname{rank}_{\mathbb{Z}_p} \operatorname{Sel}_{p^\infty}$ *denotes the corank of the* p^∞*-Selmer group. Consequently, if* $Ш(E/\mathbb{Q})[p^\infty]$ *is finite, then* $(-1)^{r_{\mathrm{alg}}} = \varepsilon(E)$.

The p-parity conjecture over $\mathbb{Q}$ was proved by Nekovář [Nek01] for p odd and good ordinary reduction and extended by Dokchitser–Dokchitser [DD10] to all p and all reduction types. Note carefully the structure of the statement: what is proved unconditionally is a statement about the *Selmer corank*; the passage to the rank of $E(\mathbb{Q})$ requires the finiteness of Ш. This pattern — unconditional control of Selmer, conditional control of Mordell–Weil — is universal, and Chapter 20 explains why it is structural rather than accidental.

1.4 The conjecture

Conjecture 1.4.1 (Birch–Swinnerton-Dyer, rank part). *[OPEN] For every elliptic curve* $E/\mathbb{Q}$*,*

$$\operatorname{ord}_{s=1} L(E,s) \;=\; \operatorname{rank} E(\mathbb{Q}).$$

Conjecture 1.4.2 (Birch–Swinnerton-Dyer, full form). *[OPEN] For every elliptic curve* $E/\mathbb{Q}$*, the group* $Ш(E/\mathbb{Q})$ *is finite,* $r := \operatorname{ord}_{s=1} L(E,s) = \operatorname{rank} E(\mathbb{Q})$*, and*

$$\lim_{s\to 1} \frac{L(E,s)}{(s-1)^r} \;=\; \frac{\#Ш(E/\mathbb{Q}) \cdot \operatorname{Reg}(E/\mathbb{Q}) \cdot \Omega_E \cdot \prod_p c_p}{\left(\#E(\mathbb{Q})_{\mathrm{tors}}\right)^2}.$$

Theorem 1.4.2 is the statement adopted by the Clay Mathematics Institute as one of its Millennium Problems, in the formulation of Wiles [Wil06]. It subsumes Theorem 1.4.1 and Theorem 1.2.4.

Remark 1.4.3 (On the shape of the formula). The right-hand side of Theorem 1.4.2 is not a random assemblage. It is, up to normalisation, a Tamagawa measure: $\Omega_E \prod_p c_p$ is the volume of $E(\mathbb{A}_{\mathbb{Q}})$ with respect to a global differential, and $\#Ш \cdot \operatorname{Reg} / (\#E(\mathbb{Q})_{\mathrm{tors}})^2$ is the correction accounting for the failure of the local–global principle and for the covolume of the Mordell–Weil lattice. This observation was made precise by Tate [Tat66] and vastly generalised by Bloch–Kato [BK90] and Fontaine–Perrin-Riou [FPR94]: the BSD formula is the case of an elliptic curve of a general conjecture expressing the leading term of the L-function of a motive as the determinant of a fundamental line. That reformulation is the historical ancestor of the framework of this book, and we treat it in detail in Chapter 14.

Remark 1.4.4 (The ℓ-parts). It is often convenient to break Theorem 1.4.2 into its ℓ-primary parts: for each prime ℓ one asks that the ℓ-adic valuations of the two sides agree. This is strictly weaker prime by prime but equivalent when taken over all ℓ together with a rationality statement. Almost all known results towards the full formula are of this ℓ-part type, and the primes $\ell = 2, 3$ and the primes of additive reduction are systematically harder. Chapter 26 records exactly which ℓ-parts are known.

1.5 What is known

We now record the principal unconditional results. They are, without exception, confined to analytic rank at most one.

Theorem 1.5.1 (Coates–Wiles). *[KNOWN] Let* $E/\mathbb{Q}$ *have complex multiplication by an order in an imaginary quadratic field of class number one. If* $L(E,1) \neq 0$ *then* $E(\mathbb{Q})$ *is finite.*

This is [CW77]; it is the first theorem in the direction of Theorem 1.4.1.

Theorem 1.5.2 (Gross–Zagier). *[KNOWN] Let $E/\mathbb{Q}$ be of conductor N and let K be an imaginary quadratic field satisfying the Heegner hypothesis for N. Let $y_K \in E(K)$ be the Heegner point. Then $L'(E/K,1)$ is a non-zero explicit multiple of the canonical height $\hat{h}_K(y_K)$. In particular $L'(E/K,1) \neq 0$ if and only if y_K is non-torsion.*

This is the theorem of Gross and Zagier [GZ86]. Its significance is that it is the *only* known mechanism for producing a rational point of infinite order from analytic information, and it produces exactly one.

Theorem 1.5.3 (Kolyvagin). *[KNOWN] Let $E/\mathbb{Q}$ be an elliptic curve with $\mathrm{ord}_{s=1}L(E,s) \leq 1$. Then*

$$\operatorname{rank} E(\mathbb{Q}) \;=\; \mathrm{ord}_{s=1}L(E,s) \qquad \text{and} \qquad \#Ш(E/\mathbb{Q}) < \infty.$$

This is Kolyvagin's theorem [Kol88; Kol90], combined with Theorem 1.5.2 and Theorem 1.3.1. It is, sixty years after the conjecture was made, still the strongest unconditional statement towards the rank part, and Chapter 25 is devoted to explaining precisely why the hypothesis $\mathrm{ord}_{s=1}L(E,s) \leq 1$ cannot be relaxed by any refinement of the method.

Theorem 1.5.4 (Kato; Skinner–Urban; and others). *[KNOWN] Let $E/\mathbb{Q}$ with $\mathrm{ord}_{s=1}L(E,s) \leq 1$ and let $p \geq 5$ be a prime of good ordinary reduction at which the mod-p Galois representation is surjective and certain ramification hypotheses hold. Then the p-part of the formula of Theorem 1.4.2 holds.*

The divisibility in one direction is Kato [Kat04]; the reverse is Skinner–Urban [SU14], with the rank-one case completed by Skinner [Ski20], W. Zhang [Zha14b] and others. The precise hypotheses are stated in Chapter 26; they are numerous, and the reader should resist the temptation to summarise this result as "the p-part of BSD is known in rank ≤ 1".

Theorem 1.5.5 (Bhargava–Shankar; Bhargava–Skinner–Zhang). *[KNOWN] When elliptic curves over $\mathbb{Q}$ are ordered by naive height, the average rank is bounded above by an absolute constant less than 1; a positive proportion of curves have rank 0 and a positive proportion have rank 1; and a positive proportion satisfy Theorem 1.4.1.*

See [BS15a; BS15b] and [BSZ14]. These are theorems about a *population*, obtained by averaging Selmer group sizes over the family; they are not, and cannot by their method be, theorems about any specified curve of rank ≥ 2.

Theorem 1.5.6 (A. Smith). *[KNOWN] In quadratic twist families, the distribution of $Ш[2^\infty]$ is as predicted by the Delaunay heuristics, and the 2-part of Goldfeld's conjecture on average ranks holds.*

See [Smi17].

The state of the art, in one sentence

Every unconditional theorem asserting $\operatorname{rank} E(\mathbb{Q}) = \mathrm{ord}_{s=1}L(E,s)$ for a *specified* curve E requires $\mathrm{ord}_{s=1}L(E,s) \leq 1$; and for no elliptic curve over $\mathbb{Q}$ of rank ≥ 2 is $Ш(E/\mathbb{Q})$ known to be finite.

1.6 Three worked examples

Example 1.6.1 (The curve $37.a1$: rank one). Let $E : y^2 + y = x^3 - x$, of conductor 37. This is the elliptic curve of smallest conductor with positive rank. One has $E(\mathbb{Q}) \cong \mathbb{Z}$, generated by

$P = (0,0)$, and $E(\mathbb{Q})_{\mathrm{tors}} = 0$. The sign of the functional equation is $\varepsilon(E) = -1$, so r_{an} is odd, and $L'(E,1) \neq 0$, whence $r_{\mathrm{an}} = 1$. All Tamagawa numbers are 1. The BSD quotient

$$\frac{L'(E,1) \cdot (\#E(\mathbb{Q})_{\mathrm{tors}})^2}{\mathrm{Reg}(E/\mathbb{Q}) \cdot \Omega_E \cdot \prod_p c_p}$$

evaluates numerically to 1, consistent with $\#Ш(E/\mathbb{Q}) = 1$, and here Ш is provably trivial by Theorem 1.5.3 together with a descent computation. This curve is therefore a case in which Theorem 1.4.2 is a theorem.

Example 1.6.2 (The curve 389.*a*1: rank two). Let $E : y^2 + y = x^3 + x^2 - 2x$, of conductor 389. This is the elliptic curve of smallest conductor with rank 2; one has $E(\mathbb{Q}) \cong \mathbb{Z}^2$ with generators $(0,0)$ and $(-1,1)$, trivial torsion, and $\varepsilon(E) = +1$. Numerically $L(E,1) = L'(E,1) = 0$ and $L''(E,1) \neq 0$, so $r_{\mathrm{an}} = 2$ — but this last assertion combines a numerical computation with a rigorous error bound showing $L''(E,1) \neq 0$ with the parity theorem; it is not obtained by pure thought.

Everything else about this curve that BSD asserts is unknown. Theorem 1.4.1 is not proved for 389.*a*1. $Ш(E/\mathbb{Q})$ is not known to be finite for 389.*a*1. No known construction produces two independent points of $E(\mathbb{Q})$ from the vanishing of L and L'. The curve 389.*a*1 is the smallest concrete object on which every technique described in Chapter 24 through Chapter 26 fails, and it will serve throughout this book as the standing test case: any proposed advance must be able to say something about 389.*a*1 that is not already known.

Example 1.6.3 (The curve 5077.*a*1: rank three). Let $E : y^2 + y = x^3 - 7x + 6$, of conductor 5077, the curve of smallest conductor with rank 3. Here $\varepsilon(E) = -1$, the torsion is trivial, and the three generators $(-2,3)$, $(-1,3)$, $(0,2)$ are independent. Again nothing beyond the numerics and the parity theorem is known. We record it because the distinction between even and odd analytic rank ≥ 2 matters for the constructions of Part IV: an odd-rank curve at least has a non-vanishing sign obstruction to exploit, whereas 389.*a*1 offers none.

1.7 What the conjecture does not say

Three clarifications will prevent misreadings later.

First, Theorem 1.4.1 is not a statement about the *size* of $E(\mathbb{Q})$, nor about the height of its generators. Even granting the conjecture in full, no effective bound on the height of a generator of $E(\mathbb{Q})$ follows; effectivity is a separate and also open question, connected to the Gross–Zagier formula via Goldfeld's theorem on class numbers.

Second, the conjecture is not a statement that can be checked for a single curve by a finite computation, in either direction. Rigorous numerics can certify $L^{(r)}(E,1) \neq 0$, hence give $r_{\mathrm{an}} \leq r$ only when combined with an independent lower bound; and computing $\mathrm{rank}\, E(\mathbb{Q})$ requires either a successful descent, which terminates only if one can rule out n-divisible elements of Ш that the descent fails to detect, or a point search, which is not guaranteed to terminate. The apparatus of Part VII is precisely an account of how far these obstructions can be pushed back.

Third, the conjecture says nothing about *why* the two sides should be equal. It is an assertion of numerical coincidence between quantities defined by unrelated procedures. The thesis of this book is that this is the essential difficulty and that it is a defect of the foundations rather than of the conjecture; Chapter 2 makes the case.

Status of this chapter

This chapter is expository. Theorem 1.2.1, Theorem 1.2.2, Theorem 1.2.3, Theorem 1.3.1, Theorem 1.5.1, Theorem 1.5.2, Theorem 1.5.3, Theorem 1.5.4, Theorem 1.5.5 and Theorem 1.5.6 are [KNOWN] with the hypotheses as stated. Theorem 1.3.3 is [KNOWN] for Selmer coranks and yields information about Mordell–Weil ranks only conditionally on finiteness of Ш. Theorem 1.2.4, Theorem 1.4.1 and Theorem 1.4.2 are [OPEN]. Nothing in this chapter is new.

Notes and References

The two papers of Birch and Swinnerton-Dyer, *Notes on elliptic curves I, II*, appeared in *J. reine angew. Math.* in 1963 and 1965; the conjecture in the form given here first appears in the second. Tate's 1966 Bourbaki report [Tat66] contains the reformulation in terms of Tamagawa measures and the observation, still the sharpest available in the function-field case, that BSD for an elliptic surface over a finite field is equivalent to the finiteness of its Brauer group; Milne [Mil75] completed that circle of ideas. Silverman [Sil09] remains the standard reference for everything in Section 1.2; Cremona [Cre97] for the computational apparatus.

The modularity theorem has an extensive literature; the reader wanting the architecture rather than the details should consult the Cornell–Silverman–Stevens volume [CSS97]. For parity, Nekovář's *Selmer complexes* [Nek06] is the systematic treatment, and Dokchitser–Dokchitser [DD10] the sharpest statement.

For the state of computational verification, the LMFDB [The26] is the authoritative record; the reader should note that "BSD verified" in that database means the full formula for curves of rank ≤ 1 subject to explicit hypotheses, and an equality of numerical approximations otherwise. Chapter 47 discusses what a genuine certificate would have to contain.

Nothing in this chapter is due to the present work. Its function is to fix the inventory: a framework must reproduce every theorem listed in Section 1.5, and must explain, rather than merely inherit, the barrier at analytic rank one.

Chapter 2

Anatomy of the Obstruction

The difficulty of the Birch and Swinnerton-Dyer conjecture lies not in either of its two sides but in the absence of a category containing both. We dissect the two sides into explicit chains of constructions, observe that the chains share only their input, and isolate four diagnoses: a category gap, an archimedean gap, a linearity gap, an integrality gap. The function-field analogue, where the conjecture is a theorem, is then examined ingredient by ingredient against its missing arithmetic counterpart. We ask of each foundational reform of the last sixty years what it fixed and what it left, and formulate the Construction Problem attempted in Parts I–IV.

2.1 Two chains with a common source

The conjecture of Chapter 1 asserts the equality of two integers and then of two real numbers. It is worth setting out how each is obtained, because the shape of the difficulty is legible in the bare list. Write $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. The analytic side is a composite

$$E \;\rightsquigarrow\; V_\ell E \;\rightsquigarrow\; f_E \;\rightsquigarrow\; L(E,s) \;\rightsquigarrow\; \mathrm{ord}_{s=1} L(E,s) \tag{2.1.1}$$

of four constructions of four quite different kinds. *First*, $E \rightsquigarrow V_\ell E := H^1_{\text{ét}}(E_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)^\vee$ is a genuine functor, to continuous ℓ-adic representations of $G_{\mathbb{Q}}$; it is exact, and it is *lossy*, factoring through the isogeny category and so blind to $E(\mathbb{Q})_{\text{tors}}$, Ω_E and $\prod_p c_p$. *Second*, $V_\ell E \rightsquigarrow f_E \in S_2(\Gamma_0(N))$ is not a functor at all: it is the modularity theorem (Chapter 31), whose output lives among automorphic representations of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$, which admit no functor to or from ℓ-adic representations. It is an identification of two families of numbers indexed by primes, not a construction of one object from another. *Third*, $f_E \rightsquigarrow L(E,s)$ produces a holomorphic function; there is no category of L-functions and no useful notion of morphism between them. *Fourth*, $L(E,s) \rightsquigarrow \mathrm{ord}_{s=1} L(E,s)$ is a limiting operation at a point outside the half-plane of convergence of the Euler product, yielding an integer that is a local invariant of a transcendental object.

The algebraic side is a different composite,

$$E \;\rightsquigarrow\; \mathcal{E}/\mathbb{Z} \;\rightsquigarrow\; \mathrm{Sel}_n(E/\mathbb{Q}) \subseteq H^1(\mathbb{Q}, E[n]) \;\rightsquigarrow\; E(\mathbb{Q}) \;\rightsquigarrow\; \mathrm{rank}\, E(\mathbb{Q}). \tag{2.1.2}$$

The Néron model $\mathcal{E}$ is a smooth commutative group scheme over $\mathrm{Spec}\mathbb{Z}$, not proper, whose bad fibres carry the component groups from which the c_p are read; the Néron functor is a right adjoint, is not exact, and does not commute with H^1. Passage to Galois cohomology is an Ext-computation for the Kummer sequence followed by local conditions at every place; its natural home is Nekovář's derived category of Selmer complexes [Nek06]. The third step *does not close*: descent computes Sel_n exactly and recovers $E(\mathbb{Q})/nE(\mathbb{Q})$ only up to $Ш[n]$. The fourth is an invariant of a finitely generated abelian group.

Observation 2.1.1. The two chains agree at their source and nowhere else. At every intermediate stage they name objects of categories — Tannakian, automorphic, derived Galois, analytic — between which no functor is known in either direction. They agree again at the end, but only because both have been reduced to a bare natural number from which all structure has been discarded.

2.2 The category gap

Principle 2.2.1 (The category gap). *[NEW] There is no category in which $L(E,s)$ and $E(\mathbb{Q})$ are both objects, and no functor is known relating them other than through modularity, which is a* comparison *of invariants, not a* construction *of one side from the other.*

The distinction is not rhetorical, and the following elementary proposition shows what it costs.

Proposition 2.2.2. *[KNOWN] Let $\phi\colon E \to E'$ be an isogeny over $\mathbb{Q}$. Then $L(E,s) = L(E',s)$ and $\operatorname{rank} E(\mathbb{Q}) = \operatorname{rank} E'(\mathbb{Q})$. By contrast the five quantities $\#Ш$, Reg, Ω, $\prod_p c_p$, $\#E(\mathbb{Q})_{\mathrm{tors}}$ are not individually isogeny invariants; that their combination is one is a theorem of Cassels* [Cas66a].

Proof. An isogeny of degree d has a dual $\widehat{\phi}$ with $\widehat{\phi}\phi = [d]$, so ϕ induces an isomorphism $V_\ell E \xrightarrow{\sim} V_\ell E'$ of $G_{\mathbb{Q}}$-modules for every ℓ, with inverse $d^{-1}V_\ell(\widehat{\phi})$; conductor and local Euler factors are determined by the family $(V_\ell)_\ell$ through inertia invariants, whence $L(E,s) = L(E',s)$. Likewise $\phi \otimes \mathbb{Q}$ is invertible on Mordell–Weil groups. The last assertion is [Cas66a]. □

Here is the gap in miniature. The analytic side is an invariant of the isogeny class; the algebraic side is not, and becomes one only after five factors are assembled into a quotient. Cassels proves that invariance by a third argument, place by place. In a framework where the two sides were trivialisations of one object it would be a formality; as things stand it is a theorem. It is also sometimes said that modularity bridges the two sides. It does not: it bridges the ℓ-adic realisation of E and an automorphic form, both on the analytic side. The theorems that touch $E(\mathbb{Q})$ consume modularity as an input and then perform a separate construction.

2.3 The archimedean gap

Principle 2.3.1 (The archimedean gap). *[NEW] The analytic side is governed by the place ∞: the completed L-function, the functional equation and the location of $s = 1$ are archimedean data. The algebraic side treats ∞ as a correction term, the real period. The difficulty over $\mathbb{Q}$ is therefore located precisely at ∞; over a function field, where there is no archimedean place, the conjecture is a theorem conditional only on a finiteness statement.*

Theorem 2.3.2 (Tate; Grothendieck; Milne; Kato–Trihan). *[KNOWN] Let $k = \mathbb{F}_q$, let X/k be a smooth projective geometrically connected curve with function field $K = k(X)$, and let E/K be an elliptic curve with relatively minimal regular proper model $\pi\colon \mathcal{E} \to X$, a surface over k; by Lang–Néron* [LN59], *$E(K)$ is finitely generated. Then:*

(*i*) $Ш(E/K) \cong \mathrm{Br}(\mathcal{E})$ [Gro68];
(*ii*) *the following are equivalent:* $\mathrm{Br}(\mathcal{E})$ *is finite; the Tate conjecture for divisors holds on $\mathcal{E}$; the rank part of BSD holds for E/K* [Tat66];
(*iii*) *if they hold, the Artin–Tate formula, hence the leading-coefficient form of BSD for E/K, follows* [Mil75];
(*iv*) *finiteness of $Ш(E/K)[\ell^\infty]$ for one prime ℓ, including $\ell = \operatorname{char} k$, suffices* [KT03].

The contrast with the number field could not be sharper. Over $\mathbb{Q}$ it is *not* known that finiteness of $Ш(E/\mathbb{Q})$ implies BSD. What is known is conditional in a second, independent way: Flach and Morin [FM18] deduce BSD from finiteness of Ш *together with* the expected structural properties of a Weil-étale complex, themselves unproved. That extra hypothesis says precisely that the arithmetic surface behaves over its compactification as $\mathcal{E}$ does over k — and the compactification is the Arakelov one, at ∞.

Remark 2.3.3. Two features of the number-field case trace back to ∞. (*i*) The factor $\Gamma_{\mathbb{C}}(s) = (2\pi)^{-s}\Gamma(s)$ in $\Lambda(E,s)$ has no Euler-product interpretation; in Deninger's programme it is a regularised determinant on an archimedean cohomology [Den94; Den98], but no such cohomology exists. (*ii*) The Néron–Tate height mixes a Green's function with intersection multiplicities, and the real period is an integral over $E(\mathbb{R})$; Arakelov theory [Ara74; Fal84] makes such terms commensurable, but not by a common construction.

2.4 The linearity gap

Principle 2.4.1 (The linearity gap). *[NEW] Every known mechanism converting analytic vanishing into arithmetic classes — Gross–Zagier, Kolyvagin, Kato, Beilinson–Flach — produces* one *class from* one *derivative. No multilinear mechanism is known. Where a d-dimensional output is required, the known constructions supply an object whose d-fold exterior power vanishes for formal reasons.*

The full diagnosis, in five clauses, is Chapter 27. Here we record the linear algebra that makes the principle sharp, and why the cyclotomic deformation supplies no further directions.

Proposition 2.4.2. *[KNOWN] Let c lie in a $\mathbb{Q}_p$-vector space of dimension $d \geq 2$. Then $\wedge^d(\mathbb{Q}_p c) = 0$. If $\langle\,,\,\rangle$ is a symmetric bilinear form on a d-dimensional real space W and $L \subseteq W$ is a line, the restriction of the form to L determines one diagonal entry of the Gram matrix and, for $d \geq 2$, nothing about its determinant. This does not say that the Néron–Tate pairing degenerates in higher rank — it is positive definite on $E(\mathbb{Q}) \otimes \mathbb{R}$; what degenerates is the* pullback *of the determinant pairing along a one-dimensional source.*

Proof. $c \wedge c = 0$, so every exterior power of a line in degree ≥ 2 vanishes. For the second statement take a basis $w_1, \dots, w_d$ with w_1 spanning L: the restriction determines $\langle w_1, w_1\rangle$ alone, while $\det(\langle w_i, w_j\rangle)$ is non-constant on the fibres of $(\langle w_i, w_j\rangle) \mapsto \langle w_1, w_1\rangle$ once $d \geq 2$. □

Lemma 2.4.3. *[KNOWN] Let $\Gamma \cong \mathbb{Z}_p$, $\Lambda = \mathbb{Z}_p[[\Gamma]]$, and let $I = \ker(\Lambda \to \mathbb{Z}_p)$ be the augmentation ideal. Then I^r/I^{r+1} is $\mathbb{Z}_p$-free of rank one for every $r \geq 0$.*

Proof. A topological generator γ and $\gamma \mapsto 1 + T$ give $\Lambda \cong \mathbb{Z}_p[[T]]$ carrying I to (T); hence $I^r/I^{r+1} = (T^r)/(T^{r+1})$ is generated by the class of T^r, and is free of rank one because $\mathbb{Z}_p[[T]]$ is a domain. □

The consequence is structural. A one-variable main conjecture records an order of vanishing r through the graded piece I^r/I^{r+1}; but that piece is a line. It records one Taylor coefficient, a scalar, and not r independent global directions.

Remark 2.4.4 (*Core rank, and the derived Hecke mechanism*). **[KNOWN]** Two further facts confirm that the line is intrinsic. (*a*) Mazur and Rubin [MR04] attach to a self-dual p-adic representation with a self-dual Selmer structure an integer, the *core rank*; when it is one, the module of Kolyvagin systems is free of rank one and a leading-term theorem computes Fitting ideals. For elliptic curves the core rank is one: the rank-one theory is not the first case of a general theory awaiting extension, but the case in which the governing module is a line. Higher-rank Euler, Kolyvagin and Stark systems, in the sense of Büyükboduk and of Burns–Sakamoto–Sano, take values in an exterior bidual of degree d, a different object for which no construction over $\mathbb{Q}$ is known. (*b*) The derived Hecke algebra of Venkatesh and the motivic action of Prasanna–Venkatesh [Ven19; GV18; PV21] predict, for an arithmetic group of defect δ, a degree-shifting action of an exterior algebra on cohomology. For $\mathrm{GL}_2/\mathbb{Q}$ in weight two, $\delta = 0$: the cohomology of a modular curve sits in a single degree and the action contributes nothing.

2.5 The integrality gap

Principle 2.5.1 (The integrality gap). *[NEW] A proof of the leading-coefficient formula up to a factor in $\overline{\mathbb{Q}}^\times$, or even in $\mathbb{Q}^\times$, would not be a proof of the formula. The Tamagawa numbers, the square of the torsion order and the ℓ-parts at small and at additive primes require integral information that no known transcendental method sees.*

Deligne's conjecture on critical values [Del79], and here the theory of modular symbols, gives $L(E,1)/\Omega_E \in \mathbb{Q}$; Gross–Zagier gives a rational multiple relating $L'(E/K,1)$ to a height. Neither computes the rational number. What is asserted is

$$\frac{L^{(r)}(E,1)}{r!\,\Omega_E\,\mathrm{Reg}(E/\mathbb{Q})} \;=\; \frac{\#Ш(E/\mathbb{Q})\cdot\prod_p c_p}{(\#E(\mathbb{Q})_{\mathrm{tors}})^2},$$

whose right-hand side is a product of integral quantities each defined by a lattice or component-group computation. Comparison of two archimedean quantities up to $\overline{\mathbb{Q}}^\times$ is blind to all of it. This is why almost every known result towards the formula is an ℓ-part statement, and why $\ell = 2, 3$ and the additive primes are hardest. A framework proposing to prove the formula must carry a lattice, not a vector space, through every step — including the archimedean step, where no lattice is at present available.

2.6 The function-field mirror

The comparison with the function field is the most illuminating material available on why BSD is hard, and must be made ingredient by ingredient.

2.6.1 What is provable there, and by what means

Keep the notation of Theorem 2.3.2. By Tate [Tat66], $L(E/K, s)$ is a polynomial in q^{-s}: a characteristic polynomial of geometric Frobenius on a piece of $H^2_{\text{ét}}(\overline{\mathcal{E}}, \mathbb{Q}_\ell)$. The Shioda–Tate formula [Shi90]

$$\rho(\mathcal{E}) \;=\; 2 + \sum_v (m_v - 1) + \operatorname{rank} E(K) \tag{2.6.1}$$

expresses the Picard number through the rank and the numbers m_v of components of bad fibres, and the Tate conjecture identifies $\rho(\mathcal{E})$ with the multiplicity of q as a Frobenius eigenvalue on H^2; the rank part is the resulting equality of two multiplicities. The Artin–Tate formula [Mil75] then expresses the special value through the order of the Brauer group, the discriminant of the intersection pairing on the Néron–Severi lattice and local terms, and with (2.6.1) becomes the leading-coefficient statement. Finally Yun–Zhang [YZ17; YZ19] prove, for suitable everywhere unramified data over K, that for every $r \geq 0$ the r-th central derivative of a base-changed L-function is the self-intersection number of a Heegner–Drinfeld cycle on the moduli stack of shtukas with r legs. Their dictionary — legs against order of derivative against codimension of a derived intersection — is the only known mechanism producing r independent geometric directions from an r-th derivative. Its absence over $\mathbb{Q}$ is Theorem 2.4.1.

2.6.2 What has no arithmetic counterpart

Each proof consumes an ingredient that does not exist over $\mathbb{Z}$.

(i) *The curve X and the constant field k.* Every construction above begins with X as a variety over a field. $\operatorname{Spec}\mathbb{Z}$ is not a variety over anything: it has no field of constants, and its "dimension one" is an analogy, not a statement about a base.

(ii) *The product $X \times_k X$ and the diagonal.* Weil's proof of the Riemann hypothesis for curves, and every trace-formula argument descended from it, lives on a self-product. But $\operatorname{Spec}\mathbb{Z} \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec}\mathbb{Z} = \operatorname{Spec}\mathbb{Z}$: there is no base beneath $\mathbb{Z}$, hence no diagonal to intersect with a graph.

(iii) *The global Frobenius* Frob_q. A single morphism of $\mathcal{E}$ defined simultaneously everywhere, whose fixed points are the k-rational points and whose eigenvalues on cohomology are the local data. Over $\mathbb{Z}$ there are only the local Frob_p, in different Galois groups.

(iv) *Properness.* $\mathcal{E} \to \operatorname{Spec} k$ is proper, so its cohomology is finite-dimensional and satisfies Poincaré duality. $\operatorname{Spec}\mathbb{Z}$ is not; the Arakelov compactification of Chapter 4 is not a scheme, and no cohomology on it is known to be finite and self-dual over $\mathbb{Z}$.

(v) *The Weil conjectures for the surface.* Deligne [Del74] supplies purity: eigenvalues of Frob_q on H^2 have absolute value q, so the multiplicity of the eigenvalue q is an isolated invariant. Over $\mathbb{Z}$ no cohomology is known whose Frobenius eigenvalues are the coefficients of $L(E, s)$; "multiplicity of an eigenvalue" has no meaning, and the order of vanishing cannot be a dimension.

(vi) *The Hitchin fibration and support theorems.* Yun–Zhang fibre moduli of shtukas over a Hitchin-type base and use support theorems of Ngô's kind [Ngô10]. That base is a space of sections of a bundle on X; neither it nor the theorem survives passage to $\mathbb{Z}$.

(vii) *Commensurability of orbit lengths.* There closed orbits have lengths in $\log q \cdot \mathbb{Z}$, and one has a $\mathbb{Z}$-action. Over $\mathbb{Z}$ the lengths would be $\log p$ for varying p, rationally independent; the $\mathbb{Z}$-action must become an $\mathbb{R}$-action, a flow. This is why the object sought in Chapter 15 is dynamical, not algebraic.

Example 2.6.1 (A constant elliptic surface, where everything works). Let $E_0/\mathbb{F}_q$ be an ordinary elliptic curve, $X/\mathbb{F}_q$ a smooth projective geometrically connected curve with a rational point, $K = \mathbb{F}_q(X)$, and $E = E_0 \times_{\mathbb{F}_q} K$, with model $\mathcal{E} = E_0 \times_{\mathbb{F}_q} X$. Fixing a base point identifies $E(K) = \operatorname{Mor}_{\mathbb{F}_q}(X, E_0) \cong E_0(\mathbb{F}_q) \oplus \operatorname{Hom}_{\mathbb{F}_q}(J_X, E_0)$, J_X the Jacobian, so $\operatorname{rank} E(K) = \operatorname{rank}_{\mathbb{Z}} \operatorname{Hom}_{\mathbb{F}_q}(J_X, E_0)$. On the other side $\operatorname{NS}(E_0 \times X) \cong \mathbb{Z}^2 \oplus \operatorname{Hom}_{\mathbb{F}_q}(J_X, E_0)$, so $\rho(\mathcal{E}) = 2 + \operatorname{rank} E(K)$: this is (2.6.1) with all $m_v = 1$. Tate's theorem on homomorphisms of abelian varieties over finite fields yields the Tate conjecture for divisors

on a product of two curves, hence $Br(\mathcal{E})$ and $Ш(E/K)$ are finite and, by Theorem 2.3.2, the full BSD formula holds for E/K.

Now take $X = E_0$. Then $\mathrm{Mor}_{\mathbb{F}_q}(E_0, E_0) = \mathrm{End}_{\mathbb{F}_q}(E_0) \oplus E_0(\mathbb{F}_q)$, and for E_0 ordinary $\mathrm{End}_{\mathbb{F}_q}(E_0)$ is an order in an imaginary quadratic field, of $\mathbb{Z}$-rank 2. Hence $\mathrm{rank}\, E(K) = 2$, $\mathrm{ord}_{s=1} L(E/K, s) = 2$, $\#Ш(E/K) < \infty$, and the leading-coefficient formula is a theorem: a case of BSD in rank two, proved. Rank two is not intrinsically beyond reach. What is beyond reach is rank two *over* $\mathbb{Q}$, and the difference is exactly the seven ingredients above.

Example 2.6.2 (The curve 389.*a*1, where nothing works). Let $E : y^2 + y = x^3 + x^2 - 2x$, of conductor 389: rank 2 with generators $(0, 0)$ and $(-1, 1)$, trivial torsion, $\varepsilon(E) = +1$ (Chapter 1). There is no curve X; the relevant surface fibres over $\mathrm{Spec}\mathbb{Z}$ and is not proper. There is no global Frobenius, so $L(E, s)$ is the characteristic polynomial of nothing. There is no Shioda–Tate formula, since the Néron–Severi group of an arithmetic surface does not contain the Mordell–Weil group in the required way without an Arakelov completion, and no completion with the required finiteness is available. Gross–Zagier applies over an imaginary quadratic field satisfying the Heegner hypothesis, but analytic rank at least 2 there forces the Heegner point y_K to be torsion, so by Theorem 2.4.2 Kolyvagin's descent has no non-torsion input. Kato's Euler system supplies a class whose 2-fold exterior power vanishes. The p-adic L-function vanishes to order at least 2, and by Theorem 2.4.3 the graded piece recording its leading term is still a line. Finally $Ш(E/\mathbb{Q})$ is not known to be finite, so even the conditional route of Flach–Morin is closed. Every clause is a known fact and none is a technical difficulty: 389.*a*1 is the smallest object on which the entire apparatus fails at once, and it is the standing test case of this book.

2.7 Why the foundational reforms did not suffice

Each reform below is a major achievement, and the point is never that it failed but that it addressed a different gap. The semicolon separates what was fixed from what was left.

Schemes and topoi.
Geometry over any base, with nilpotents and descent; but a locally ringed space over $\mathbb{Z}$ has no Frobenius, no fibre at ∞ and no regularised determinant.

Étale cohomology and the Weil conjectures.
A cohomology whose Frobenius eigenvalues are the local zeta factors [Del74]; but only over a field, so over $\mathbb{Z}$ it needs a global Frobenius.

Motives and the standard conjectures.
A language in which $h^1(E)$ carries an L-function; but the standard conjectures [Gro69] are open, and $h^1(E)$ knows only the isogeny class.

Arakelov geometry.
Intersection theory and Riemann–Roch including the fibre at ∞ [Ara74; GS90; GS92]; but that fibre is a Green's function, not a space, so there is no cohomology of the compactified object and no determinant of a complex.

Iwasawa theory.
p-adic L-functions against characteristic ideals, with control of ℓ-parts [MW84; Rub91; Kat04; SU14]; but the deformation is one-dimensional and ∞ is absent altogether.

p-adic Hodge theory.
The étale–de Rham comparison and the local conditions defining H^1_f [Sch13]; but one prime at a time, with no coherence across p and no archimedean member.

Condensed and analytic geometry.
Topological and algebraic structures in one category, with nuclearity and traces [CS19; CS20]; but a substrate with no arithmetic content: no Frobenius, flow or Selmer group.

Prismatic cohomology and F-gauges.
Integral p-adic cohomology unifying crystalline, de Rham and étale theories with Frobenius [BS22; Dri20; Bha22]; but one prime at a time, and no prism at ∞.

The ninth entry needs a paragraph. The Bloch–Kato and Fontaine–Perrin-Riou determinant formalism [BK90; FPR94; BF01] fixed the correct general shape of the statement: the leading term of the L-function of a motive is the comparison of two structures on a single *fundamental line*. This is by a

wide margin the closest antecedent to the present framework, and the debt is total. What it left is that the formalism *reformulates*: the equivariant Tamagawa number conjecture for $h^1(E)(1)$ is equivalent to BSD, not easier, and the two trivialisations are still built by unrelated procedures. *The same criticism applies to this book.* It is Obstruction VIII, pressed against the framework in Chapter 53.

Observation 2.7.1. The pattern is uniform. Each reform supplied one structure that the function-field proof uses; none supplied several simultaneously on one object. The four gaps above are not gaps within any of these theories; they are gaps between them.

2.8 The Construction Problem, and what would count as progress

Problem 2.8.1 (The Construction Problem). [OPEN] Construct a category $\mathbf{C}$, an object $X_E \in \mathbf{C}$ functorially attached to $E/\mathbb{Q}$, a cohomology theory $\mathrm{R}\Gamma_{\mathrm{FAP}}(X_E, -)$ with values in a symmetric monoidal stable ∞-category admitting regularised determinants, and a flow Φ^t on that cohomology with generator Θ, such that:

(i) the regularised determinant of $s - \Theta$ on $\mathrm{R}\Gamma_{\mathrm{FAP}}(X_E, \mathbb{M}_E)$ is $\Lambda(E, s)$, the archimedean Γ-factor arising as the determinant of a boundary contribution;

(ii) the fixed locus of the flow — the fibre of $\Theta - 1$ — is the Selmer complex of E, with induced Bockstein the Néron–Tate height pairing;

(iii) consequently the determinant line $\Delta(X_E) = \det \mathrm{R}\Gamma_{\mathrm{FAP}}(X_E, \mathbb{M}_E)$ carries two trivialisations, λ_{an} from (i) and λ_{alg} from (ii), whose equality is the conjecture.

The precise form of this problem, with the specification F1–F12 that a solution must meet, is in Chapter 7 and Chapter 8. It is not solved in this book, and the statement "its solution, with one further conjecture, implies BSD" has value only if the left-hand side is a better target than the right. Eight obstructions stand between formulation and solution (Chapter 53): **I** the adelic structured ∞-topos — no site is known carrying prismatic, analytic and archimedean covers at once; **II** coherence of prismatic gauges across all p; **III** the archimedean place, where there is no prism; **IV** the arithmetic flow — no $\mathbb{R}_{\geq 0}$-action with closed orbits of length $\log p$ has ever been constructed; **V** nuclearity and continuation, the regularised determinant requiring trace-class behaviour and a heat expansion nobody has verified; **VI** independence of realisation; **VII** the Selmer fixed-point principle, clause (ii) carrying nearly all the arithmetic content; **VIII** equality of the two trivialisations, integrally of ETNC strength.

Remark 2.8.2. Obstructions IV and VII are at least as hard as major open problems in their own right: IV subsumes the substance of Deninger's programme [Den94; Den98], and VII subsumes the finiteness of Ш. A reader who concludes from Theorem 2.8.1 that BSD has been reduced to something easy has read it wrongly.

2.8.1 What would count as progress, and what would refute the approach

A programme that cannot say what would falsify it is not a programme; both lists are made precise in Chapter 55.

Genuine advance. (i) A hyperdescent theorem for the adelic topology of Chapter 9, with excision for $\mathbb{G}_m$ over $\mathbb{Z}$. (ii) Recovery of the analytic class number formula for $\mathbb{G}_m/\mathbb{Z}$, archimedean term included, from the trace formula of Chapter 16: the minimal test of whether the archimedean gap has been closed. (iii) Recovery of the Grothendieck–Lefschetz formula, and of Theorem 2.3.2, from that formula. (iv) A prismatic formula for c_p as a Lefschetz–Verdier local term, checkable against Tate's algorithm [Tat75]. (v) An integral multi-leg norm relation at $r = 2$, with congruences verified on 389.a1. (vi) A construction of two independent classes in $H^1_f(\mathbb{Q}, V_pE)$ from the vanishing of L and L' for one curve of rank two.

Refutation of the approach. (i) An inconsistency proof for F1–F12, or a demonstration that its only models are the two already known. (ii) A no-go theorem for Obstruction IV: a proof that no $\mathbb{R}_{\geq 0}$-action of the required kind exists on any cohomology of $\mathrm{Spec}\mathbb{Z}$. (iii) A disagreement between the prismatic formula for c_p and Tate's algorithm on one curve. (iv) A proof that nuclearity of the critical complex is strictly weaker or stronger than finiteness of Ш.

Items (iii)–(v) of the first list and (iii)–(iv) of the second are testable now, independently of BSD. That, not the conjecture, is the near-term claim of this book.

Status of this chapter

[NEW] Theorem 2.2.1, Theorem 2.3.1, Theorem 2.4.1 and Theorem 2.5.1 are formulations, not assertions of new mathematical fact; each is supported by a statement proved here or cited. [KNOWN] Theorem 2.2.2; Theorem 2.3.2; Theorem 2.4.2 and Theorem 2.4.3, proved here; Theorem 2.4.4. [OPEN] Theorem 2.8.1, with Obstructions I–VIII. **Confidence.** High: the chapter is assembled from known theorems and elementary linear algebra, and its novelty is the assembly. **Would be refuted by** a functorial construction relating $L(E, s)$ and $E(\mathbb{Q})$ without an intervening comparison, or a proof of BSD in rank ≥ 2 using only rank-one inputs.

Notes and References

The diagnosis of Section 2.4 is assembled rather than new; its five clauses are proved in Chapter 27. The core-rank formalism is [MR04]; the higher-rank theory alluded to in Theorem 2.4.4 is due to Büyükboduk and to Burns, Sakamoto and Sano. For Section 2.6 see [Tat66], [Gro68], [Mil75], [KT03], [Shi90], [LN59], and for the higher-derivative theory [YZ17; YZ19], on [Dri87; Laf18], with support technology from [Ngô10]. For the dynamical antecedents see [Den94; Den98; Con99; CC16], and for the Weil-étale determinant formalism [Lic05; FM18; FM21]; what this book takes from each is in Chapter 3. A last remark on register: having listed seven ingredients the function-field proof uses and the arithmetic case lacks, one might conclude that supplying them is a programme. It is not. It is a list of what nobody knows how to supply, and its only value is that it is finite and explicit.

Chapter 3

The Spectral Descent Programme

This chapter states the programme of the book: what it constructs, what it conjectures, what would follow if the conjectures held, and — at equal length — what it does not do. The governing idea is to replace the comparison of two numbers by the construction of one object with two descriptions. We set out the three routes to that object, the way they depend on one another, the logical spine running from the foundational axioms to the classical statements, and the register of twelve conjectures on which everything rests. Eleven of the twelve are graded four or five stars. The book does not prove the conjecture of Birch and Swinnerton-Dyer, establishes no new case of it, and proves the finiteness of Ш *for no curve.*

3.1 One object with two descriptions

The diagnosis of Chapter 2 was that the two sides are manufactured in categories with no common home, and that every passage between them — modularity above all — compares separately constructed things rather than constructing one thing. The response attempted here is stated once and never elaborated.

> *Replace the comparison of two numbers by the construction of a single object admitting two descriptions, and prove that the descriptions agree.*

This is not a new idea; it is the idea of the Lefschetz trace formula, of Tamagawa measure, and of the equivariant Tamagawa number conjecture. What is proposed is a particular realisation of it for $h^1(E)(1)$ over $\mathbb{Q}$, with three moving parts and two operations on them.

(i) *A space.* An object X_E — the *BSD arithmetic spectrum* of Chapter 18 — built over the *FAP spaces* of Chapter 7, in which the prismatic, the analytic and the archimedean local geometries sit side by side rather than in separate theories. $\mathrm{Spec}\,\mathbb{Z}$ is not such a substrate; the four gaps of Chapter 2 are the four ways in which it fails to be one.

(ii) *A cohomology.* A theory $\mathrm{R}\Gamma_{\mathrm{FAP}}(X_E, -)$ valued in a category where trace and determinant are defined — nuclear objects in the condensed sense — so that finiteness becomes functional-analytic rather than combinatorial.

(iii) *A flow.* A one-parameter action Φ^t, $t \in \mathbb{R}_{\geq 0}$, with generator Θ, whose closed orbits are the primes, the orbit at p having length $\log p$, and whose time-$\log p$ map is the Frobenius at p. This is the object Deninger has sought since 1991 [Den91; Den94; Den98].

The two operations are a *spectral* one and a *descent* one, whence the name of the programme. The spectral operation is the regularised determinant: if Θ acts on $H^i_{\mathrm{FAP}}(X_E, \mathbb{M}_E)$ with a spectrum sparse enough for zeta-regularisation, one may form

$$Z_E(s) \;:=\; \prod_i \det_\infty\Big(\tfrac{s-\Theta}{2\pi} \,\Big|\, H^i_{\mathrm{FAP}}(X_E, \mathbb{M}_E)\Big)^{(-1)^{i+1}}, \tag{3.1.1}$$

and the first half of the programme is the assertion that $Z_E(s) = \Lambda(E, s)$, Γ-factor included. The point is not the formula — modularity already gives one — but its *kind*: the archimedean factor is a determinant over an archimedean orbit rather than a convention imported from Hodge theory [Del79]. This is the only device in the book addressing the archimedean gap directly.

The descent operation is the passage to the fixed locus. Classical descent computes $E(\mathbb{Q})$ as the invariants of a Galois action; here one takes the fibre of $\Theta - 1$ on the same complex, and the assertion is that this fibre is the Selmer complex of Nekovář [Nek06], with the Bockstein of the flow equal to the Néron–Tate height pairing. The rank then appears not as the size of a group but as the multiplicity of the eigenvalue 1 of Θ, that is, as an *index*.

Both descriptions trivialise one graded line, the *fundamental line* $\Delta(X_E)$ of Chapter 37: the spectral one, λ_{an}, from the leading behaviour of (3.1.1) at $s = 1$; the descent one, λ_{alg}, from the determinant of the fixed-point complex together with its local terms. The conjecture of Birch and Swinnerton-Dyer, in this framework, is the statement that they coincide, integrally. That is the Master Identity, `conj:master-identity`.

Observation 3.1.1. What this buys is precise and limited: a *single well-posed target* in place of an equality between two unrelated computations, a place where the archimedean factor is not a correction, and a rank formula of index type. It does not buy a proof. The equality of two trivialisations of a determinant line is exactly the difficulty Bloch and Kato isolated [BK90], and constructing the line does not make it easier. This is Obstruction VIII.

3.2 Three routes, and how they depend on one another

The programme is pursued along three routes. They are not independent assaults on the same wall; two are inputs to the third, and their compatibility is itself an unproved conjecture. Each is stated in four lines — constructs, would give, needs, stops at — and then the dependencies.

Route 1: the spectral route (Parts I–III and VI)

Constructs: the FAP substrate, the twelve axioms F1–F12, the flow Φ^t, the categorified trace formula, the spectrum X_E and its critical complex, the fundamental line and its two trivialisations. *Would give:* the rank formula as an index theorem, and the full leading-coefficient formula, with the archimedean place on the same footing as the finite ones. *Needs:* a solution of the Construction Problem of Chapter 7, consistency of F1–F12, and Conjectures A, G, K, L, M. *Stops at:* Obstruction IV. No $\mathbb{R}_{\geq 0}$-action with closed orbits of length $\log p$ has ever been constructed on any cohomology over $\mathrm{Spec}\mathbb{Z}$. The flow is postulated, and the only models of F1–F12 exhibited in Chapter 8 are $\mathbb{G}_m/\mathbb{Z}$ and curves over $\mathbb{F}_q$ — the two cases where the answer was known beforehand.

Route 2: the shtuka and Euler-system route (Parts III–IV)

Constructs: a moduli problem $\mathrm{Sht}^{(r)}_{\mathbb{Z}}$ of arithmetic shtukas with r legs over a prismatic adelic base (Chapter 29), a class $\kappa^{(r)}$ in an exterior bidual of Selmer cohomology, and multi-leg norm relations producing one Euler factor per leg. *Would give:* the rank lower bound $\mathrm{rank}\, E(\mathbb{Q}) \geq r$ when $L^{(r)}(E,1) \neq 0$, and, with Conjecture F together with [Kat04; SU14], the p-part of the leading coefficient in the good ordinary regime. *Needs:* Conjectures B and F, and the archimedean non-vanishing supplied by Route 3. *Stops at:* Obstructions IX, X and XII. There is no global Frobenius on $\mathrm{Spec}\mathbb{Z}$, so the closing isomorphism defining a shtuka has no known meaning; the integral multi-leg norm relations are unproved over any number field; and the rank *upper* bound needs a derived rank-r descent for which Kolyvagin's derivative machinery has no analogue.

Route 3: the automorphic and theta route (Part V)

Constructs: an r-legged cycle class $Z^{(r)}_E$ on an arithmetic Shimura-type object, in the tradition of Gross–Zagier [GZ86], the Kudla programme [Kud97; KRY06] and the arithmetic fundamental lemma [Zha21; LZ22a]. *Would give:* the archimedean identity $\langle Z^{(r)}_E, Z^{(r)}_E \rangle = L^{(r)}(E,1)/(r!\,\Omega_E)$ up to explicit local factors — the non-vanishing input without which Route 2 relates two objects that may both be zero. *Needs:* modularity [Wil95; TW95; BCDT01] and a number-field analogue of [YZ17; YZ19]. *Stops at:* Obstruction XI. The Yun–Zhang proof is a statement about the geometry of a curve, a Hitchin base and a proper moduli stack. Over $\mathbb{Q}$ there is no curve, no Hitchin base and no properness.

The relations

(i) *Route 3 is an input to Route 2, not a parallel to it.* The archimedean clause of Conjecture B *is* the conclusion of Conjecture C. A reciprocity law between a class and a p-adic L-function yields a rank bound only when some period is known non-zero; that is the service Gross–Zagier performs for Kolyvagin at $r = 1$ [Kol90; Gro91]. Without C, Conjecture B is a bookkeeping identity. The residual implication is Conjecture E.

(ii) *Neither Route 1 nor Route 2 implies the other.* Conjecture A is a statement about the determinant of a complex; Conjecture B is a statement about a class. A together with D implies the cohomological clause of B; but A does *not* follow from B and C even given D and E, because A asserts an integral identity for the whole complex, including H^0, H^3 and the bad fibres, which no Euler system sees.

(iii) *The spectral route is not logically necessary.* B, F, H and J together yield the numerical BSD formula with no flow at all. Route 1's claim is therefore explanatory and structural — that the rank is an index, and that ∞ becomes a place like the others — not a claim of logical indispensability. Chapter 41 draws this.

(iv) *The compatibility of the routes is itself conjectural.* That the leg filtration of Route 2 agrees with the Θ-Bockstein filtration of Route 1 is Conjecture D, graded five stars. Until it is proved the three routes are three constructions hoped to describe one object.

Warning 3.2.1. There is a fourth route, and omitting it would be dishonest. Flach and Morin [FM18; FM21], on Lichtenbaum's Weil-étale cohomology [Lic05], prove that *if* $Ш(E/\mathbb{Q})$ is finite and the Weil-étale complex has its expected properties, *then* BSD follows from a special-value formula (Chapter 14). The present framework therefore cannot be judged by whether it implies BSD — so does that package, with far less machinery. It can be judged only by whether the Construction Problem is more tractable than the finiteness of Ш. We do not know that it is.

3.3 The spine

The following is the dependency chain in prose, each link tagged. **[NEW]** marks a formulation introduced here, never an assertion of truth; **[CONDITIONAL]** marks a statement proved subject to earlier links; **[KNOWN]** marks imported mathematics.

(1) **The Construction Problem** (Chapter 7) **[OPEN]**. Construct a substrate over which the FAP formalism satisfies F1–F12.

(2) **The axioms F1–F12** (Chapter 8) **[NEW]**. A specification, not a theorem. They are not known to be consistent; two models are exhibited, $\mathbb{G}_m/\mathbb{Z}$ and curves over $\mathbb{F}_q$.

(3) **Six operations, descent, realisations** (Chapter 9, Chapter 10) **[CONDITIONAL]** on F1–F2 and F10. Hyperdescent for the adelic topology — the first checkable technical claim in the book — and ℓ-independence of local terms (Obstruction VI).

(4) **The arithmetic flow** (Chapter 15) **[OPEN]**, and equal to Conjecture L. Existence, uniqueness, and the identification of the time-log p map with Frobenius, via the cyclotomic structure on topological Hochschild homology [NS18].

(5) **The categorified trace formula** (Chapter 16) **[CONDITIONAL]** on F1–F12. The trace of Φ^t decomposes into prime-orbit local terms and an archimedean counterterm: Weil's explicit formula [Wei52], categorified.

(6) **Regularised determinants** (Chapter 17) **[CONDITIONAL]** on F9 and Conjecture M: (3.1.1) is defined and equals $\Lambda(E, s)$.

(7) **The spectrum, the critical complex and the local terms** (Chapter 18, Chapter 20, Chapter 22) **[NEW]**. Axiom F12 identifies the fixed locus of the flow with the Selmer complex and the flow Bockstein with the Néron–Tate pairing — Obstruction VII, where all the arithmetic content is concentrated — and c_p becomes a Lefschetz–Verdier local term, falsifiable against Tate's algorithm [Tat75].

(8) **Finiteness** (Chapter 23) **[OPEN]**, and equal to Conjecture K. Finiteness of Ш becomes nuclearity of the critical complex.

(9) **The fundamental line and the Master Identity** (Chapter 37) **[NEW]**, Conjecture A.

(10) **The rank formula** (Chapter 38) **[CONDITIONAL]** on (1) and A: $\operatorname{rank} E(\mathbb{Q}) = \operatorname{ord}_{s=1} L(E, s)$, as an index formula.

(11) **The leading coefficient** (Chapter 39) **[CONDITIONAL]** on (1), A, finiteness of Ш, and Conjectures H and J.

Links (1), (4), (8) and (9) are the load-bearing ones, and all four are open. The chain is therefore not a proof under weakening hypotheses; it is a decomposition of one hard problem into a specific list of hard problems whose individual statements can be attacked, tested or refuted separately. That is the whole of the claim.

3.4 The twelve conjectures

Twelve statements carry the programme; they are stated formally in their own chapters. Difficulty is on a five-point scale: $\star$ a good exercise with the framework in hand; $\star\star$ a paper; $\star\star\star$ a hard paper or a thesis; $\star\star\star\star$ a research programme; $\star\star\star\star\star$ subsumes or exceeds a known major open problem. The letter I is not used, to avoid collision with Obstruction I.

A (Ch. 37, $\star\star\star\star\star$), `conj:master-identity`.
The two trivialisations of $\Delta(X_E)$ agree integrally. Implies the rank formula and the leading coefficient.

B (Ch. 30, $\star\star\star\star\star$), `conj:higher-reciprocity`.
An r-variable reciprocity law relates $\kappa^{(r)}$ to a plectic p-adic L-function, with archimedean period $L^{(r)}(E,1)/(r!\,\Omega_E)$. Implies the rank lower bound.

C (Ch. 35, $\star\star\star\star\star$), `conj:higher-agz`.
An r-legged cycle over $\mathbb{Q}$ has derived self-intersection $L^{(r)}(E,1)/(r!\,\Omega_E)$. Implies the archimedean clause of B.

D (Ch. 37, $\star\star\star\star\star$), `conj:spectral-shtuka`.
The leg filtration equals the Θ-Bockstein filtration. Joins Route 1 to Route 2 in both directions.

E (Ch. 35, $\star\star\star\star$), `conj:orbit-period`.
The archimedean regularisation constant equals the period normalisation of C. Pins the 2π and the Γ-factor.

F (Ch. 30, $\star\star\star\star\star$), `conj:plectic-mc`.
A plectic derived main conjecture. With B, gives the p-part of the leading coefficient.

G (Ch. 20, $\star\star\star\star\star$), `conj:graded-vanishing`.
The critical complex carries a free graded action whose rank is $\mathrm{ord}_{s=1}L$ and whose top piece is the regulator. The only candidate mechanism for $r \geq 2$ inside Route 1.

H (Ch. 39, $\star\star\star$), `conj:integral-hasse`.
Local equality at every p and at ∞ implies global integral equality. Assembles prime-by-prime results.

J (Ch. 39, $\star\star\star\star$), `conj:integral-orientation`.
One $\mathbb{Z}$-lattice orientation of $\Delta(X_E)$ realises correctly at every p, including 2, 3 and additive primes. With H, upgrades all p-parts to BSD.

K (Ch. 23, $\star\star\star\star\star$), `conj:nuclear-sha`.
The critical complex is nuclear if and only if $Ш(E/\mathbb{Q})$ is finite. Recasts an old problem as a new kind of statement.

L (Ch. 15, $\star\star\star\star\star$), `conj:lambda-flow`.
The flow exists, is unique, and restricts to Frobenius at $t \in \log p \cdot \mathbb{Z}_{\geq 0}$. Everything in Route 1 rests on it.

M (Ch. 17, $\star\star\star\star$), `conj:heat-kernel`.
The trace of Φ^t is polyhomogeneous as $t \to 0^+$, so $\det_\infty$ is defined and $\mathcal{Z}_E$ continues meromorphically.

Remark 3.4.1 (*The ratio*). Eleven of the twelve are graded four or five stars, and only H is tractable with present technique. Four — A, C, K, L — subsume or exceed a known major open problem: A is the equivariant Tamagawa number conjecture for $h^1(E)(1)$ in other clothing; C is a number-field Yun–Zhang; K is the finiteness of Ш; L is Deninger's programme in full. A framework that replaces one conjecture by twelve, of which eleven are harder than a thesis, has bought structure at a price, and the price must be quoted openly.

Standing conjectures are cited but never renumbered: finiteness of Ш; the equivariant Tamagawa number conjecture [BK90; FPR94; BF01]; the Iwasawa main conjecture [Kat04; SU14]; the standard conjectures [Gro69]. These are inputs, never contributions. Alongside the conjectures runs the atlas of fourteen obstructions, I–VIII blocking the Construction Problem and IX–XIV the arithmetic and analytic routes, catalogued with attack plans in Chapter 53; six practical obstructions to computation and formalisation are separate, in Chapter 55.

3.5 What the programme claims, and what it does not

This is the paragraph the reader is entitled to have without ornament.

This book does not prove the conjecture of Birch and Swinnerton-Dyer. It does not prove any new case of it. It does not prove the finiteness of $Ш(E/\mathbb{Q})$ *for a single elliptic curve. It does not prove that the axioms F1–F12 are consistent, and the only structures satisfying them that we can exhibit are two in which the arithmetic conclusions were already theorems. The central object of Part IV, the stack of arithmetic shtukas with r legs, is not constructed; its defining datum is a global Frobenius on* $\mathrm{Spec}\mathbb{Z}$*, whose absence is the reason the problem is hard. Conjecture A, if proved, would be a proof of BSD; it is also, in substance, the equivariant Tamagawa number conjecture, and we have no reason to think our formulation of it is easier than the received one.*

What the book does offer is five things.

(i) *A framework* in which the analytic and algebraic sides of BSD are two trivialisations of one determinant line attached to one object, so that the conjecture becomes an identity within a single category rather than an equality across two.

(ii) *A precise chain of conjectures* reducing BSD to that framework: twelve statements, each with a label, a chapter, a difficulty grade and a list of what it implies and depends on, so that the reduction can be audited and any link attacked in isolation.

(iii) *A diagnosis of the rank-two barrier* sharper than what was previously written down. Chapter 27 assembles five independent reasons why every classical Euler-system input is a one-dimensional object, and isolates the mechanism: the cyclotomic deformation records a scalar, the relevant graded piece of the Iwasawa algebra is free of rank one, and therefore no refinement of cyclotomic Iwasawa theory can produce r independent tangent directions.

(iv) *An atlas of fourteen obstructions*, each with what it blocks, a difficulty grade, a named attack plan, and an explicit statement of whether it subsumes a known open problem. Four of the fourteen do.

(v) *Infrastructure*: an engine producing certificates with declared result levels rather than assertions (Chapter 47), and a Lean 4 development (Chapter 48) which records that Néron models, Tate's algorithm, canonical heights, Selmer groups and modularity are all absent from the formal libraries.

We claim nothing beyond these five. We do not claim that the framework is close to completion, that any obstruction is close to removal, or that the programme is more likely to succeed than the alternatives of Section 3.6. A reader who finishes the book convinced only that the four gaps of Chapter 2 are the right four gaps will have taken from it everything we are confident is there.

3.6 Antecedents

The framework is an assembly; almost every component belongs to someone else, and the parts that do not are the joints. We record the debts.

Deninger's dynamical programme.
[Den91; Den94; Den98]. That Λ should be a regularised determinant of a flow generator, that $\Gamma_{\mathbb{C}}(s)^{-1}$ is $\det_\infty((s-\Theta)/2\pi)$, and that primes are closed orbits of length $\log p$, is his, unchanged. *Taken:* (3.1.1) and Conjecture L. *Added:* a proposed substrate and a specification for what the missing space must satisfy, so that the programme has a target.

Connes' spectral approach.
[Con99; CC16]. The trace formula as organising principle, the scaling site as a model of an archimedean flow. *Taken:* the shape of the archimedean counterterm. *Added:* nothing; we use it as template and test.

Bloch–Kato and Fontaine–Perrin-Riou.
[BK90; FPR94; BF01]. The determinant-line formulation of special values, the conditions H^1_f, and the notion of a fundamental line are theirs; Conjecture A is their conjecture for $h^1(E)(1)$ in another category. *Added:* the flow, hence a mechanism by which the archimedean trivialisation is computed rather than defined. Whether that is progress is what Chapter 55 makes testable.

Weil-étale cohomology.
[Lic05; FM18; FM21]. Lichtenbaum's insight that a Weil group gives Euler characteristics of the

right shape, and Flach–Morin's construction and special-value theorem, are the closest existing realisation of "one object, two descriptions". *Taken:* the fundamental-line formalism of Chapter 14 and the benchmark of Theorem 3.2.1. *Added:* a continuous flow where they have a discrete Weil group — a hypothesis, not an improvement.

The Kudla programme.
[Kud97; KRY06]. Generating series of arithmetic cycles whose heights are derivatives of Eisenstein series. *Taken:* the template for Conjecture C. *Added:* nothing; Part V is an exposition with a conjecture at the end, and Chapter 35 says so.

Yun–Zhang.
[YZ17; YZ19], on [Dri87] and [Laf18]. The identification "r legs = r-th derivative" is theirs, and is the single most important structural idea in Part IV. *Added:* a specification of what an arithmetic analogue must satisfy, and a demonstration (Obstruction XI) that the proof does not transport.

Venkatesh's derived structures.
[Ven19; PV21]. A graded action explaining higher-dimensional cohomology. *Taken:* the vocabulary of Conjecture G. *Not taken:* the mechanism, which does not apply — the defect of $\mathrm{GL}_2/\mathbb{Q}$ is zero, so the derived Hecke algebra is trivial in exactly our case. Any reader who takes Part II to supply a mechanism for $r \geq 2$ has been misled, and Chapter 53 states this as Obstruction XIV.

The novelty of this book, such as it is, lies in the assembly and in the diagnosis: in insisting that these programmes are addressing four separable gaps, in specifying what an object closing all four would have to satisfy, and in grading the distance to it. We claim priority for no idea above.

3.7 Leitfaden

The book has ten parts. Part 0 (Chapters 1–3) is the argument; Part I (4–10) the foundations; Part II (11–17) higher structure, flow and traces; Part III (18–23) the spectrum X_E; Part IV (24–30) Euler systems and shtukas; Part V (31–36) the automorphic input; Part VI (37–41) the Master Identity and its descent to the classical statements; Part VII (42–47) computation; Part VIII (48–52) formalisation; Part IX (53–56) the accounting. The hard dependencies are I → II, III; II, III → IV; III → V; IV, V → VI; VI → VII, VIII; VII, VIII → IX.

Three paths are recommended. *The foundational reader* takes Chapters 1–17, then 37, then 53–54: the geometry, the axioms, the flow, the trace formula, the Master Identity, and what remains unbuilt. *The arithmetic reader* takes Chapters 1–3, 18–30, then 38 and 41, consulting Parts I–II rather than reading them; the centre of gravity is Chapter 27, independent of the new machinery. *The computational reader* takes Chapters 1–3 and 42–52, then 55, a path independent of Parts I–VI and the one to take for what is testable.

3.8 Closing assessment

A hostile referee would say the following, and most of it would be right. One conjecture has been replaced by twelve axioms and twelve conjectures, of which eleven are graded four or five stars and not one is easier than what it replaces. Axiom F12 is BSD. Conjecture A is the equivariant Tamagawa number conjecture. The Construction Problem has no solution in any case where the answer was not already known, so there is no evidence that the axioms are consistent. The central higher-rank object requires a Frobenius on $\mathrm{Spec}\mathbb{Z}$; the book names a lemma for it and moves on. The engine verifies the numerical BSD ratio, which Cremona's tables have done for decades. And the one sharp result, the rank-two diagnosis, is assembled linear algebra together with Mazur–Rubin core rank [MR04]: correctly put together, but not new.

We do not dispute this. The reply is not an argument but four experiments, each independent of BSD and each capable of failing in a locatable place. *Stage A*: the trace formula of Chapter 16 must reproduce the analytic class number formula for $\mathbb{G}_m/\mathbb{Z}$, archimedean term included. *Stage B*: it must reproduce Grothendieck–Lefschetz, and with it the function-field case of BSD [Tat66; Mil75]. *Stage C*: the prismatic formula for c_p must agree with Tate's algorithm [Tat75] on every curve of small conductor. *Stage D*: the predicted integral multi-leg norm relations at $r = 2$ must hold as congruences

between classes on 389.$a1$. If all four succeed, the framework has earned a hearing; if one fails, it is wrong somewhere specific. That, and not BSD, is the near-term claim of this book.

It is worth saying what the expectation is. The three-star obstructions — the adelic site, prismatic coherence across all p, independence of realisation — may well be removed by ordinary hard work within a decade. The five-star ones will not be removed by this book or by any of its methods, because each subsumes a problem the subject has failed to solve for thirty years or more. The framework relocates the difficulty of BSD; it does not reduce it. Its value, if it has any, is that the relocation is precise enough to be checked, and that the fourteen places where it fails are now written down with names.

Status of this chapter

[NEW] Nothing is proved here. The route decomposition, the spine, the register of Conjectures A–M and the atlas of fourteen obstructions are formulations, stated formally in the chapters cited. [KNOWN] All cited results. [OPEN] The Construction Problem; Conjectures A–M; Obstructions I–XIV. **Confidence.** High that the decomposition is faithful to the later chapters; low that the programme will be completed. **Would be refuted by** a proof of BSD making no use of a single object with two descriptions; a demonstration that F1–F12 are inconsistent; or failure of any of Stages A–D above.

Notes and References

The dynamical programme is Deninger's [Den91; Den94; Den98; Den08]; the spectral realisation of the explicit formula [Wei52] is Connes' [Con99; CC16]. Higher-rank Euler system formalism is in [MR04; Rub00] and [BSS21]; plectic structures are Nekovář and Scholl's [NS16]; for the shtuka picture add [Laf02; Ngô10]. The substrates are [CS19; CS20] and [BS22; BL22b]. Statistical evidence, a falsification surface and not an input, is in [BS15b; Zub23]. A final remark on register. The list of twelve conjectures and fourteen obstructions may read as a programme in the optimistic sense. It is not one. It is an inventory of what is missing, compiled so that the missing pieces can be named, and its chief merit is that it is finite.

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I — Problem and Foundations
The anatomy of the problem; FAP spaces, the twelve axioms F1–F12, six operations, descent and traces.

II — The Arithmetic Spectrum and the Rank-Two Barrier
The spectrum of an elliptic curve, Selmer as a fixed locus, Euler systems, the rank-two diagnosis.

III — The Master Identity and the Engine
Conjecture A, the descent to the rank and leading-coefficient formulas, a certified computational engine.

IV — Formalisation and the Programme
The proof-obligation ledger, fourteen obstructions, nine stages, and what would refute the framework.

C — Companion Code · BSDKit · Lean
The Python reference implementation with its tests, and the Lean 4 sources, published uncompiled.

A — Appendices · A–H
Elliptic curves, Galois cohomology, ∞-categories, the condensed and prismatic toolkits, ledger, tables.